

BCAS2 promotes primitive hematopoiesis by sequestering β -catenin within the nucleus

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eLife Assessment

This **important** work supports the role of breast carcinoma amplified sequence 2 (Bcas2) in positively regulating primitive wave hematopoiesis through amplification of beta-catenin-dependent (canonical) Wnt signaling. The study is **convincing**: it uses appropriate and validated methodology in line with the current state-of-the-art, and there is a first-rate analysis of a strong phenotype with highly supportive mechanistic data. The findings shed light on the controversial question of whether, when, and how canonical Wnt signaling may be involved in hematopoietic development. The work will be of interest to hematologists and developmental biologists.

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Abstract

Breast carcinoma amplified sequence 2 (BCAS2), a core component of the hPrP19 complex, plays crucial roles in various physiological and pathological processes. However, whether BCAS2 has functions other than being a key RNA-splicing regulator within the nucleus remains unknown. Here, we show that BCAS2 is essential for primitive hematopoiesis in zebrafish and mouse embryos. The activation of Wnt/ β -catenin signal, which is required for hematopoietic progenitor differentiation, is significantly decreased upon depletion of *bcas2* in zebrafish embryos and mouse embryonic fibroblasts (MEFs). Interestingly, BCAS2 deficiency has no obvious impact on the splicing efficiency of β -catenin pre-mRNA, while significantly attenuating β -catenin nuclear accumulation. Moreover, we find that BCAS2 directly binds to β -catenin via its coiled-coil domains, thereby sequestering β -catenin within the nucleus. Thus, our results uncover a previously unknown function of BCAS2 in promoting Wnt signaling by enhancing β -catenin nuclear retention during primitive hematopoiesis.

Introduction

Hematopoiesis refers to the lifelong process by which all blood cell lineages are generated. It begins at the early stage of embryonic development, providing the growing embryo with sufficient oxygen and nutrients.¹ Evolutionarily conserved across vertebrate species, hematopoiesis consists of two successive and partially overlapping waves: primitive and definitive. In mammals, the first wave of hematopoiesis occurs in the yolk-sac blood islands, producing primitive erythroid, megakaryocyte and macrophage progenitors, which can be observed in mouse embryos as early as embryonic day 7.25 (E7.25).^{2–4} In zebrafish, primitive hematopoiesis initiates at around 11 hours post fertilization (hpf), when hemangioblasts emerge from the anterior lateral mesoderm (ALM) and posterior lateral mesoderm (PLM) and later differentiate into both hematopoietic and endothelial cells.^{5–7}

Breast cancer amplified sequence 2 (BCAS2), also known as pre-mRNA splicing factor SPF27, is a 26-kDa nuclear protein containing two coiled-coil (CC) domains.⁸ It was initially found to be overexpressed and amplified in human breast cancer cell lines.^{9–11} Further studies have identified BCAS2 as a vital component of the human Prp19/CDC5L complex, which forms the catalytic ribonucleoprotein (RNP) core of spliceosome and is required for the activation of pre-mRNA splicing.^{9,12,13} In *Drosophila*, the function of BCAS2 in RNA splicing is essential for cell viability.¹⁴ In mouse, disruption of *Bcas2* in male germ cells impairs mRNA splicing and leads to a failure of spermatogenesis.¹⁵ Additionally, BCAS2 has been shown to be a negative regulator of p53 by directly interacting with p53 or modulating alternative splicing of *Mdm4*, a major p53 inhibitor.^{8,16} Zebrafish *bcas2* transcripts were enriched in the sites of both primitive and definitive hematopoiesis during embryonic development.¹⁶ However, a previous study showed that p53 overactivation induced by zebrafish *bcas2* depletion did not affect primitive hematopoiesis, but impaired definitive hematopoiesis.¹⁶ In recent years, several studies have highlighted the importance of regulating the expression and activity of p53 in primitive erythroid cell differentiation in both mouse and zebrafish embryos.^{17–19} Thus, it is necessary to reexamine the exact function of BCAS2 in primitive hematopoiesis.

Wnt signaling, usually categorized into canonical and non-canonical pathways, is involved in the process of hematopoiesis.^{20–22} Notably, the canonical Wnt signaling pathway, which is dependent on the nuclear accumulation of β -catenin to regulate gene transcription, controls primitive hematopoietic progenitor formation and promotes definitive hematopoietic stem cell (HSC) specification.^{23–25} For instance, it has been demonstrated in *Xenopus* that, Wnt4-mediated activation of Wnt/ β -catenin signaling plays a critical role in the induction and maintenance of primitive hematopoiesis.²⁶ Moreover, transient inhibition of canonical Wnt signaling in zebrafish embryos impairs embryonic blood formation.²⁷ However, previous studies utilizing human pluripotent stem cells revealed an opposite role of Wnt/ β -catenin pathway in primitive progenitor generation.^{25,28} Therefore, the impact of Wnt/ β -catenin signaling on primitive hematopoiesis remains elusive and even controversial. Moreover, it has been suggested that BCAS2 is important for neural stem cell proliferation and dendrite growth in mice by regulating β -catenin pre-mRNA splicing.^{29,30} As a nuclear protein, it is unclear whether BCAS2 can modulate Wnt/ β -catenin signaling in a splicing-independent manner.

In this study, we generated two zebrafish *bcas2* mutant lines, both of which exhibited defects in male fertility and embryonic HSC formation, similar to what was previously reported in mice and zebrafish.^{15,16} More importantly, loss-of-function experiments suggest that BCAS2 is necessary for primitive hematopoiesis in both zebrafish and mouse embryos. We further find that *bcas2* is dispensable for the survival and proliferation of hematopoietic cells, but plays a crucial role in the differentiation of the hematopoietic lineage from hemangioblasts. Using a comprehensive approach, we reveal that BCAS2 is a nuclear retention factor for β -catenin during primitive

hematopoiesis. Subsequent biochemical and functional experiments demonstrate that BCAS2 directly binds to β -catenin and suppresses its nuclear export to promote Wnt signal activation and hematopoietic progenitor differentiation. Furthermore, the CC domains on BCAS2 and the Armadillo (ARM) repeats on β -catenin are responsible for their interaction. Collectively, we have uncovered a novel function of BCAS2 in regulating Wnt/ β -catenin signaling by sequestering β -catenin within the nucleus during primitive hematopoiesis.

Methods

Animal models

Our studies including animal maintenance and experiments were performed in compliance with the guidelines of the Animal Care and Use Committee of the South China University of Technology (Permission Number: 2023092). Seven strains of zebrafish were used in this study, including Tübingen wild-type, *bcas2* mutant, *cloche* mutant, *Tg(gata1:GFP)*, *Tg(coro1a:eGFP)*, *Tg(kdr1:GFP)*, and *Tg(hsp70l:dkk1b-GFP)*. *cloche* mutant, *Tg(gata1:GFP)* and *Tg(kdr1:GFP)* lines were provided by Professor Feng Liu (Chinese Academy of Sciences). *Tg(coro1a:eGFP)* was provided by Professor Yiyue Zhang (South China University of Technology). *Tg(hsp70l:dkk1b-GFP)* strain was purchased from China Zebrafish Resource Center. *Bcas2^{Floxed/Floxed}* (*Bcas2^{F/F}*) mouse line was generated as previously described.¹⁵ Genotyping of *Bcas2^{F/F}* mouse and *Flk1*-Cre mouse was performed using primers listed in **Supplemental Table 1**. The mouse model with *Bcas2* specifically disrupted in the hemangioblasts was derived from mating female *Bcas2^{F/F}* mice with *Flk1*-Cre transgenic mice. All mouse lines were maintained on a mixed background (129/C57BL/6).

Cell lines and transfection

HEK293T, HeLa, SW480, and L cell lines were obtained from ATCC. Conditional *Bcas2* knockout (*Bcas2*-cKO) mouse embryonic fibroblasts (MEFs) were prepared from *Bcas2^{F/F}* embryos at E13.5. Cells were cultured in Dulbecco's modified eagle's medium (DMEM, HyClone) supplemented with 10% fetal bovine serum (FBS, HyClone) and 1% penicillin-streptomycin (HyClone) at 37 °C and 5% CO₂. L cell lines expressing Wnt3a were maintained under similar conditions in the presence of 400 μ g/ml G-418, from which Wnt3a conditioned medium (Wnt3a CM) was generated. Culture medium prepared from L-cells was used as a control. To stimulate Wnt signaling, cells were treated with Wnt3a CM in a 1:1 ratio with normal media. To deplete *Bcas2* expression, *Bcas2*-cKO MEFs were cultured in medium containing 2 μ M tamoxifen for 72 h and the knockout efficiency was evaluated using western blot analysis. The same cells cultured without tamoxifen were used as a control. To silence BCAS2 expression, shRNA constructs in pLL 3.7-GFP plasmid were generated to target the following sequences: shRNA1, GAATGTGTAAACAATTCTA; shRNA2: GAAGGAACTTCAGAAGTTA. Transfection was performed with Lipofectamine 2000 (11668019, Invitrogen) according to the manufacturer's instructions.

Generation of CRISPR-Cas9-mediated *bcas2* knockout zebrafish

The *bcas2* knockout zebrafish mutants were generated by CRISPR-Cas9 system as previously described.³¹ The guide RNA was designed to target the sequences 5'-GGCGCAGCTGGAGCATCAGG-3' within exon 4 of *bcas2*. Humanized Cas9 mRNA and gRNA were co-injected into wild-type embryos at the 1-cell stage. Embryos or adult fin clips were collected to prepare genomic DNA. To screen for mutant alleles, the genomic regions containing gRNA-targeted sequences were amplified by polymerase chain reaction (PCR) with primers listed in **Supplemental Table 1**. The PCR products were sequenced or digested with T7 endonuclease or restriction enzyme FspI for genotyping.

RNA, morpholinos, and microinjection

Capped mRNAs for human *BCAS2*, *BCAS2* Δ CC1-2, *BCAS2* CC1-2 and mouse Δ N- β -catenin mRNA were synthesized from the corresponding linearized plasmids using an mMESSAGE mMACHINE T7 transcription kit (AM1344, Ambion). Morpholinos (MOs) were designed and purchased from Gene Tools: mismatch MO (cMO 5'-AGCCACTCATCCTGCTCCTCCCATC-3'), and *bcas2* translation-blocking MO (tMO; 5'-AGCGACTGATGCTGGTCCTGCCATC-3'). The mRNAs and morpholinos were injected into embryos at the 1- to 2-cell stage.

Whole-mount *in situ* hybridization

Digoxigenin-labeled and fluorescein-labeled probes were synthesized using a RNA Labeling kit (11175025910, Roche). Whole-mount *in situ* hybridization (WISH) and double fluorescence *in situ* hybridization for zebrafish embryos were performed following previously published methods.³²[33](#) Anti-digoxigenin-POD (11633716001, Roche) and anti-fluorescein-POD (11426346910, Roche) were used to detect digoxigenin-labeled probes and fluorescein-labeled probes, respectively. After WISH, the stained embryos were embedded in OCT and sections were prepared with a LEICA CM1900. The mouse yolk sac layers were separated as previously described.³⁴[35](#)

o-Dianisidine staining

To evaluate hemoglobin level, embryos were harvested at 36 hpf or 48 hpf, then stained with *o*-dianisidine as previously described.³⁵[36](#)

Proliferation and apoptosis assays

Embryos were incubated with 10 mM bromodeoxyuridine (BrdU) (B5002, Sigma) for 20 min. The incorporated BrdU was detected with anti-BrdU (B2531, Sigma) antibody. TUNEL staining was performed using *In Situ* Cell Death Detection Kit, TMR red (12156792910, Roche) according to the manufacturer's recommendation.

Heat shock treatment

To induce *dkk1* expression, *Tg(hsp70l:dkk1b-GFP)* embryos were subjected to heat shock (42°C) for 10 min at 10 hpf, and then collected at the indicated stage for WISH.

Dual reporter assay

HEK293T cells or MEFs were seeded in 24-well plates and transfected with a Super-TOPflash plasmid containing multimerized TCF-binding elements and a Renilla luciferase plasmid, along with the indicated vectors. Then cells were treated with 100 ng/mL LiCl and/or Wnt3a CM for 12 h and assayed for luciferase activity using the Dual luciferase system (E1910, Promega).

Immunoprecipitation, GST pulldown, and western blotting

For immunoprecipitation, HEK293T cells were transfected with the indicated plasmids and collected 48 h after transfection. Subsequently, HEK293T cells were lysed in a lysis buffer (10 mM Tris-HCl, pH 7.5, 150 mM NaCl, 2 mM EDTA and 0.5% Nonidet P-40) containing protease inhibitors. Immunoprecipitation was performed in accordance with the standard protocols.

For GST pulldown assay, GST, GST tagged β -catenin ARM 1-12 and His tagged BCAS2 were expressed in *Escherichia coli* BL21, then purified using Glutathione-Sepharose 4B beads (71024800-GE, GE Healthcare) and HisPur Ni-NTA beads (88831, Thermo Fisher), respectively. GST and GST- β -

catenin ARM 1-12 proteins were immobilized onto Glutathione-Sepharose 4B beads and incubated with purified His-BCAS2 at 4°C for 4 h. Beads were washed three times and analyzed using western blotting.

Cytoplasmic and nuclear extracts were separated with nuclear and cytoplasmic extraction kit (CW0199, CWBIO). Cell lysates were subjected to immunoprecipitation with anti-Flag M2 affinity gel (A2220, Sigma) or anti-c-Myc agarose affinity gel (A7470, Sigma) antibodies. Proteins were analyzed by western blot using anti-Flag (F2555, Sigma), anti-HA (CW0092A, CWBIO), anti- β -catenin (M24002, Abmart), anti-BCAS2 (10414-1-AP, Proteintech), anti- β -Tubulin (CW0265A, CWBIO), anti-GFP (A11120, Thermo Fisher), anti-Histone H3 (ab1791, Abcam), anti-GST (SAB4200237, Sigma), and anti-His Tag (AF5060, Beyotime) antibodies.

Immunofluorescence staining

Cells on coverslips and embryos were processed for immunofluorescence staining as previously described.^{36,37} Before fixation, *bcas2*-deficient MEFs were treated with a concentration of 20 μ M MG132 or 20 nM leptomycin B (LMB) for 6 h, while *Tg(gata1:GFP)* embryos were treated with 20 nM LMB from the bud stage to the 10-somite stage. The prepared samples were stained with anti-BCAS2 (10414-1-AP, Proteintech), anti- β -catenin (M24002, Abmart), and anti-GFP (A11122, Invitrogen). Meanwhile, 4',6-Diamidino-2'-phenylindole dihydrochloride (DAPI, 10236276001, Sigma) was used to label nuclei. Fluorescence imaging was performed using a Nikon A1R+ confocal microscope and all images were captured with the same settings. The relative fluorescence intensity was calculated by dividing the fluorescence intensity of nuclear β -catenin by the fluorescence intensity of DAPI.

Bimolecular fluorescence complementation assay

To construct the plasmids for Bimolecular fluorescence complementation assay (BiFC), BCAS2 was fused to the N-terminal half of yellow fluorescent protein (YN-BCAS2) and β -catenin to the C-terminal half (YC- β -catenin). YN-BCAS2 and YC- β -catenin were either individually or collectively transfected into HeLa cells. Fluorescence was detected 48 h after transfection using a Nikon A1R+ confocal microscope.

Fluorescence recovery after photobleaching

BCAS2 and GFP tagged S37A- β -catenin were co-transfected into HeLa cells. Fluorescence recovery after photobleaching (FRAP) assay was performed according to previously reported methods.³⁸ The cells were bleached by the 488 nm laser line of the 20 mW argon laser at 100% power. About 90% of nuclear or cytoplasmic GFP signal was bleached. Images were acquired with 35 frames at 25-s intervals by a META Zeiss 510 confocal microscope.

RNA sequencing

Embryos were collected at the 10-somite stage and gently transferred into lysis buffer. Reverse transcription was performed using a SMARTer Ultra Low RNA Kit (Clontech, 634437) directly from the cell lysates. The cDNA library was prepared using an Advantage 2 PCR Kit (Clontech, 639206) and then sequenced via the Illumina sequencing platform (NovaSeq 6000). The difference in the number of alternative splicing events between groups was analyzed using rMATS (version 4.1.0).

Reverse transcription PCR

Total RNA was isolated from wild-type and *bcas2* mutant embryos at the 10-somite stage with MicroElute Total RNA kit (OMEGA, R6831-01), followed by reverse transcription using ReverTra Ace qPCR RT Kit (Toyobo, FsQ-101). The cDNA was amplified with primers listed in **Supplemental Table 2**.

Quantification and statistical analysis

Images were quantified with Image J. Statistical data were analyzed using Graph Pad Prism. Comparisons between experimental groups were done using the Student's *t*-test. Data are presented as mean $\pm$ standard deviation. $P < 0.05$, $P < 0.01$, $P < 0.001$, and $P < 0.0001$ were considered statistically significant and marked with *, **, ***, and ****, respectively (Student's *t*-test).

Results

BCAS2 is necessary for primitive hematopoiesis

To confirm that *bcas2* is expressed in the posterior intermediate cell mass (ICM) where primitive hematopoiesis occurs in zebrafish, we first examined the spatiotemporal expression pattern of *bcas2* during zebrafish embryo development by performing whole mount *in situ* hybridization (WISH). The results showed that *bcas2* was ubiquitously expressed from 1-cell stage to 10-somite stage (**Supplemental Figure 1A** [↗](#)). Its expression in the ICM became detectable at 18 hours post fertilization (hpf) and was significantly elevated at 22 hpf (**Figure 1A** [↗](#)). We further observed that *bcas2* was co-expressed with the primitive erythropoietic marker *gata1* in the ICM at 22 hpf by fluorescence *in situ* hybridization (FISH) (**Figure 1B** [↗](#)). In contrast, *bcas2* was hardly detectable in the ICM in *cloche*^{-/-} mutants that lack both endothelial and hematopoietic cells (**Figure 1C** [↗](#)). These results demonstrate a dynamic expression of *bcas2* in the ICM and imply a potential role of this gene in primitive hematopoiesis.

To gain insight into the developmental function of *bcas2*, we employed CRISPR/Cas9 system to generate *bcas2* mutants. Two mutant lines were obtained, designated *bcas2*^{Δ7} (with a 7-base deletion) and *bcas2*^{Δ14} (with a 14-base deletion). These mutations led to premature translation termination, which resulted in truncated Bcas2 proteins lacking the C-terminal CC domains (**Supplemental Figure 2A-B** [↗](#)). *bcas2*^{+/-Δ7} and *bcas2*^{+/-Δ14} mutants were identified by T7 endonuclease I assay or restriction enzyme analysis (FspI) (**Supplemental Figure 2C** [↗](#)). We found that nearly 85% of the embryos derived from crossing *bcas2*^{+/-} males with *bcas2*^{+/-} females did not develop to the cleavage stage (**Supplemental Figure 2D** [↗](#)). Only 3% of the living embryos were homozygotes. In contrast, embryos obtained by crossing between wild-type males and *bcas2*^{+/-} females were viable and showed normal morphology, with a heterozygosity rate consistent with Mendelian inheritance. This could be explained by male infertility as previously documented in *Bcas2* knockout mice.¹⁵ [↗](#) Combining the above findings, we propose that Bcas2 may have an evolutionarily conserved role in spermatogenesis.

Given the difficulty of obtaining homozygous mutants, embryos lacking one copy of *bcas2* gene were produced from crosses between heterozygous females and wild-type males. We observed a significant decrease of *bcas2* expression in the ICM region in *bcas2*^{+/-Δ7} or *bcas2*^{+/-Δ14} mutants, likely resulting from nonsense-mediated RNA decay (**Figure 1D** [↗](#)). In line with a previous report,¹⁶ [↗](#) a marked reduction in the expression of the HSC marker *cmyb* and T-cell marker *rag1* was found in *bcas2*^{+/-Δ7} or *bcas2*^{+/-Δ14} embryos at 5 dpf, indicating an essential role of *bcas2* in definitive hematopoiesis (**Supplemental Figure 3A-B** [↗](#)). These findings suggest that *bcas2*^{+/-Δ7} and *bcas2*^{+/-Δ14} mutants can be used to examine the involvement of *bcas2* in primitive hematopoiesis.

To explore whether *bcas2* is required for primitive hematopoiesis, we first examined the expression of primitive erythropoietic markers *gata1* and *hbbe3* in *bcas2*^{+/-Δ7} and *bcas2*^{+/-Δ14} embryos at 22 hpf, and observed a marked decrease in the expression of these genes in the mutants (**Figure 1E-F** [↗](#)). Surprisingly, *o*-dianisidine staining showed similar hemoglobin contents in the *bcas2*^{+/-Δ7} and *bcas2*^{+/-Δ14} embryos at 48 hpf compared with control embryos, suggesting that the defect in primitive hematopoiesis induced by haploinsufficiency of *bcas2* was alleviated at

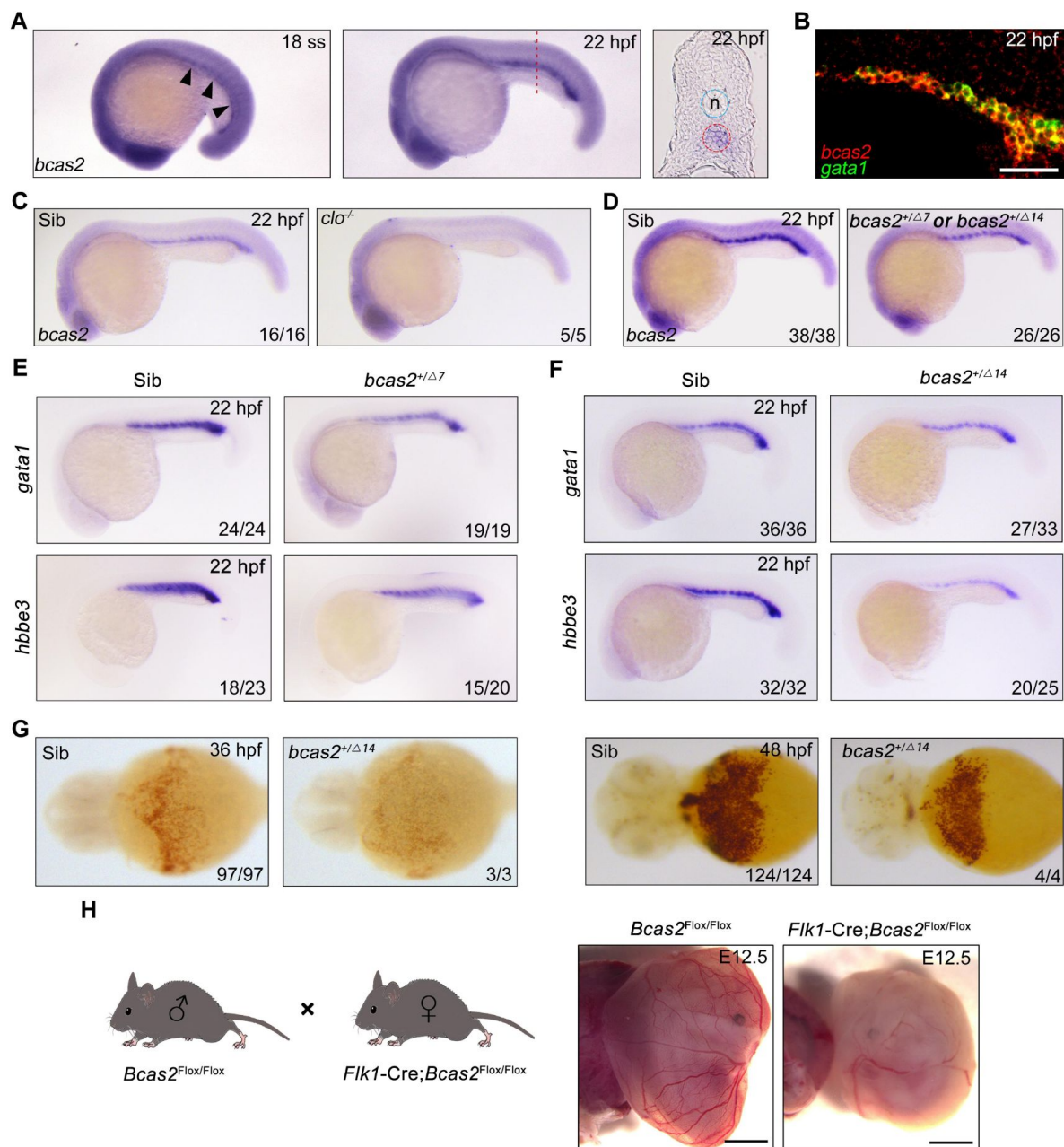


Figure 1.

***bcas2* is expressed in the ICM and required for primitive hematopoiesis.**

(A) WISH assay showing *bcas2* expression in the ICM at the 18-somite stage and 22 hpf. The dotted lines represent the section position and the black arrowheads indicate the ICM region. n, notochord. (B) Double FISH assay showing the expression pattern of *bcas2* and *gata1* in the ICM at 22 hpf. Scale bar, 50 μ m. (C-D) Comparison of *bcas2* expression in *cloche* mutants (C) or *bcas2* heterozygous mutants (D) along with their corresponding siblings. (E-F) Expression analysis of *gata1* and *hbbe3* in *bcas2*^{+/ Δ 7} and *bcas2*^{+/ Δ 14} embryos. (G) Hemoglobin detection using o-dianisidine staining in *bcas2* homozygous mutant at 36 and 48 hpf. (H) Representative images of yolk sac from the hemangioblast-specific *Bcas2* knockout mice and their siblings. *Bcas2*^{F/F} females were crossed with *Bcas2*^{F/+}; *Flk1-Cre* males to induce the deletion of *Bcas2* in hemangioblasts. Scale bar, 1 mm.

later developmental stages. In order to further explore the role of BCAS2 in primitive hematopoiesis, we identified several *bcas2*^{Δ14/Δ14} mutants from about 100 embryos. These homozygous mutants display a severe decrease in hemoglobin (**Figure 1G**). Moreover, injection of a translation-blocking MO into wild-type embryos to downregulate *bcas2* expression resulted in severe defects in erythropoiesis at 22 hpf and 48 hpf (**Supplemental Figure 4A-D**). These results indicate that *bcas2* is indispensable for primitive hematopoiesis in zebrafish. In addition, transgenic mice expressing Cre recombinase under the control of the *Flk1* promoter were crossed to *Bcas2*^{F/F} animals to induce the deletion of *Bcas2* in endothelial/hematopoietic cells. We found that red blood cells were eliminated in the yolk sac of *Bcas2*^{F/F}; *Flk1-Cre* mice at E12.5 despite the presence of vessels (**Figure 1H**). Therefore, *Bcas2* has a conserved role in vertebrates to regulate primitive hematopoiesis.

***bcas2* deficiency impairs hematopoietic progenitor differentiation**

The decrease of primitive hematopoietic cells in *bcas2* deficient animals may be attributed to a number of possible causes: excessive apoptosis, hampered proliferation of hematopoietic cells, or impaired differentiation of hematopoietic progenitor cells. To shed light on this issue, we first performed terminal deoxynucleotidyl transferase-mediated dUTP nick end labeling (TUNEL) assay in *Tg(gata1:GFP)* embryos at the 10-somite stage to examine DNA fragmentations in apoptotic cells, and found no obvious apoptotic signal in the *gata1*⁺ hematopoietic cells in either *bcas2*^{+Δ14} embryos or their wild-type siblings (**Supplemental Figure 5A**). Meanwhile, BrdU incorporation assay revealed no significant difference in hematopoietic cell proliferation between *bcas2*^{+Δ14} mutants and their corresponding wild-types (**Supplemental Figure 5B-C**). These data suggest that *bcas2* is dispensable for the survival and proliferation of hematopoietic cells.

In the developing embryo, hemangioblasts are derived from the ventral mesoderm at early somite stage, and then differentiate into both hematopoietic and endothelial lineages.^{39,40} To test whether *bcas2* functions in cell fate decision during primitive hematopoietic cell development, the expression of hemangioblast markers *npas4l*, *scl*, and *gata2* in *bcas2*^{+Δ14} embryos was examined at the 1- to 2-somite stage. As shown in **Figure 2A**, haploinsufficiency of *bcas2* did not affect the emergence of the hemangioblast population. Then we extended our analysis to include the markers of hematopoietic and endothelial progenitors. Consistent with the decrease in primitive hematopoietic cells in *bcas2* deficient mutants, a marked reduced expression of erythrocyte progenitor markers *gata1* and *hbbe3* was observed in the posterior lateral mesoderm of *bcas2*^{+/-} embryos at the 10-somite stage (**Figure 2B**). Interestingly, the expression of myeloid progenitor marker *pu.1* was also dramatically decreased (**Figure 2C**). Moreover, overexpression of human BCAS2 enhanced the expression of *gata1* in both wild-type and mutant embryos at the 10-somite stage (**Figure 2D**). In contrast, the endothelial progenitor marker *fli1a* was expressed at a similar level in *bcas2*^{+/-} embryos as in wild-type animals (**Figure 2E**). Consistently, blood vessels appeared normal in *bcas2*^{+/-} mutants with *Tg(kdrl:GFP)* background at 54 hpf (**Figure 2F**). These data provide convincing evidence that *bcas2* is required for the differentiation of the hematopoietic lineage from hemangioblasts during primitive hematopoiesis.

BCAS2 functions in primitive hematopoiesis by activating Wnt signaling

Previous studies have shown that Wnt/β-catenin plays a key role in primitive hematopoiesis.^{26,27,41} As both BCAS2 and β-catenin-like 1 (CTNBL1) are members of the Prp19/CDC5L complex, which is a major building block of the spliceosome's catalytic RNP core,¹³ we speculate that BCAS2 may be a regulator of Wnt signaling through interaction with β-catenin during hematopoiesis. To test our hypothesis, human BCAS2 was overexpressed in HEK293T cells and MEFs. Ectopic expression of BCAS2 enhanced the Wnt3a-induced expression of the TOPflash luciferase reporter in a dose-dependent manner (**Figure 3A-B**). Importantly, Wnt3a-induced luciferase activity in HEK293T cells could be effectively reduced by knockdown of *BCAS2* using

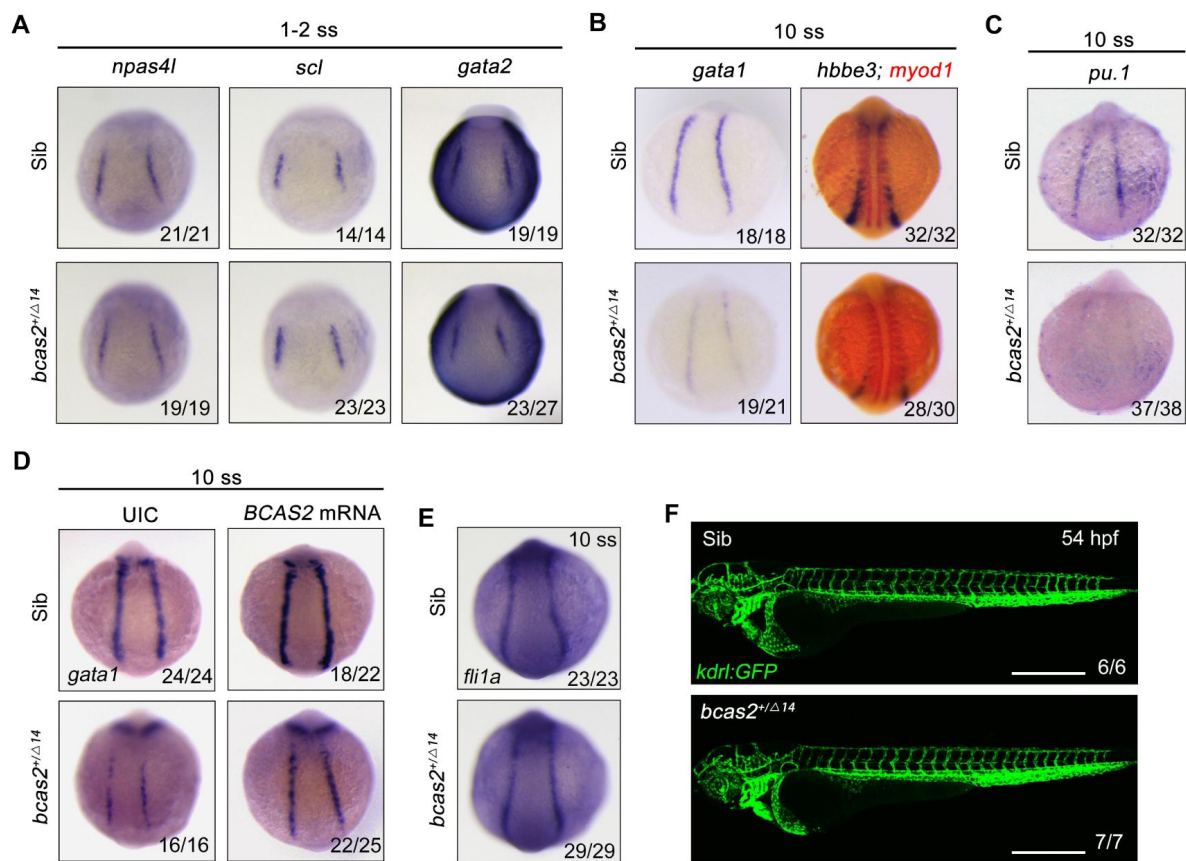


Figure 2.

***bcas2* is required for hematopoietic progenitor differentiation.**

(A-C) Expression analysis of hemangioblast markers *npas4l*, *scl*, *gata2* (A), erythroid progenitor markers *gata1*, *hbbe3* (B), and myeloid marker *pu.1* (C) in *bcas2*^{+/-Δ14} embryos and their wild-type siblings at indicated stages. (D) Expression changes of *gata1* in *bcas2*^{+/-Δ14} embryos overexpressing BCAS2 at the 10-somite stage. The indicated embryos were injected with or without 300 pg of human BCAS2 mRNA at the one-cell stage. (E) Expression of endothelial marker *fli1a* in *bcas2*^{+/-Δ14} and sibling embryos at the 10-somite stage. (G) Confocal imaging of *bcas2*^{+/-Δ14} and control sibling *Tg(kdr1:GFP)* embryos at 54 hpf. Scale bar, 500 μm.

two shRNAs targeting different regions of human *BCAS2* (**Figure 3C** and **Supplemental Figure 6**). Similar results were also observed in *Bcas2*-cKO MEFs in the presence of tamoxifen (**Figure 3D**). Furthermore, the expression of *cdx4* and *hoxa9a*, which are targets of canonical Wnt signaling in the regulation of hematopoietic development,^{42,43} were downregulated in the lateral plate mesoderm of *bcas2*^{+/-} embryos at the 6-somite stage (**Supplemental Figure 7**). These findings support that BCAS2 promotes Wnt signaling activation.

To confirm that Wnt signaling was required for zebrafish embryonic hematopoiesis, we induced the expression of canonical Wnt inhibitor Dkk1 by heat-shocking *Tg(hsp70l:dkk1-GFP)*^{w32} embryos at the bud stage.⁴⁴ As expected, diminished expression of *gata1* and *hbbe3* was detected in the resulting embryos at the 10-somite stage (**Figure 3E-F**). In addition, treatment with a small molecule β -catenin antagonist CCT036477 from 9 hpf did not affect the expression of hemangioblast markers *npas4l*,⁴⁵ *scl*, and *gata2* or endothelial marker *fli1a* (**Supplemental Figure 8A-B**), yet significantly reduced the expression of erythroid progenitor marker *gata1* in wild-type embryos (**Supplemental Figure 8C**), suggesting that canonical Wnt signaling may not be required for the generation of hemangioblasts or their endothelial differentiation, but is pivotal for their hematopoietic differentiation. To further validate that *bcas2* functions in primitive hematopoiesis via Wnt/ β -catenin signaling, the expression pattern of β -catenin was examined in *bcas2* morphants with *Tg(gata1:GFP)* background at the 10-somite stage by immunofluorescent staining. The signal of nuclear β -catenin was substantially decreased in hematopoietic progenitor cells (**Figure 3G-H**) and primitive myeloid cells (**Supplemental Figure 9A-B**). Moreover, overexpression of ΔN - β -catenin, a constitutively active form of β -catenin, effectively restored the expression of *hbbe3* in *bcas2* morphants and mutants (**Figure 3I-J**). All these data suggest that BCAS2 functions in primitive hematopoiesis by regulating Wnt/ β -catenin signaling.

BCAS2 promotes β -catenin nuclear accumulation independently of protein stability regulation

To investigate how BCAS2 regulates Wnt/ β -catenin signaling, HEK293T cells were treated with LiCl, a canonical Wnt agonist that inhibits GSK-3 β activity and stabilizes cytosolic β -catenin.⁴⁶ The results showed that TOPflash activity was significantly elevated in LiCl-treated cells (**Figure 4A**). BCAS2 overexpression further upregulated, whereas shRNA-mediated knockdown of *BCAS2* downregulated LiCl-induced TOPflash activity (**Figure 4A-B**). Likewise, HEK293T cells transfected with S37A- β -catenin, a constitutively active form of β -catenin that is resistant to GSK-3 β -mediated degradation,⁴⁷ displayed a much higher level of TOPflash activity, which was reduced by *BCAS2* knockdown (**Figure 4C**). These results strongly imply that BCAS2 regulates Wnt signaling downstream of β -catenin stability control.

To test the above hypothesis, we evaluated nuclear β -catenin level by performing immunofluorescence staining and immunoblotting experiments. Upon tamoxifen exposure, nuclear accumulation of β -catenin induced by LiCl was greatly inhibited in *Bcas2*-cKO MEFs, while nuclear/cytoplasmic fractionation suggested that cytoplasmic β -catenin level remained relatively unchanged (**Figure 4D-E**). Similarly, silencing *BCAS2* with shRNA led to reduced nuclear β -catenin in the human colon cancer cell line SW480, in which β -catenin was activated because of mutations in the adenomatous polyposis coli protein (APC), an integral component of the β -catenin destruction complex (**Figure 4F**).⁴⁸ Next, MG132, a proteasome inhibitor, was applied to activate Wnt/ β -catenin signaling in *Bcas2*-cKO MEFs by inhibiting β -catenin degradation. In the absence of tamoxifen and MG132, endogenous β -catenin was localized almost exclusively in the cytoplasm; MG132 treatment dramatically triggered β -catenin accumulation in the nuclei (**Figure 4G**). However, in *Bcas2*-cKO MEFs exposed to tamoxifen, MG132 treatment was not able to induce nuclear accumulation of β -catenin (**Figure 4G**). These findings indicate that BCAS2 promotes β -catenin nuclear accumulation in a manner that is independent of β -catenin stability regulation.

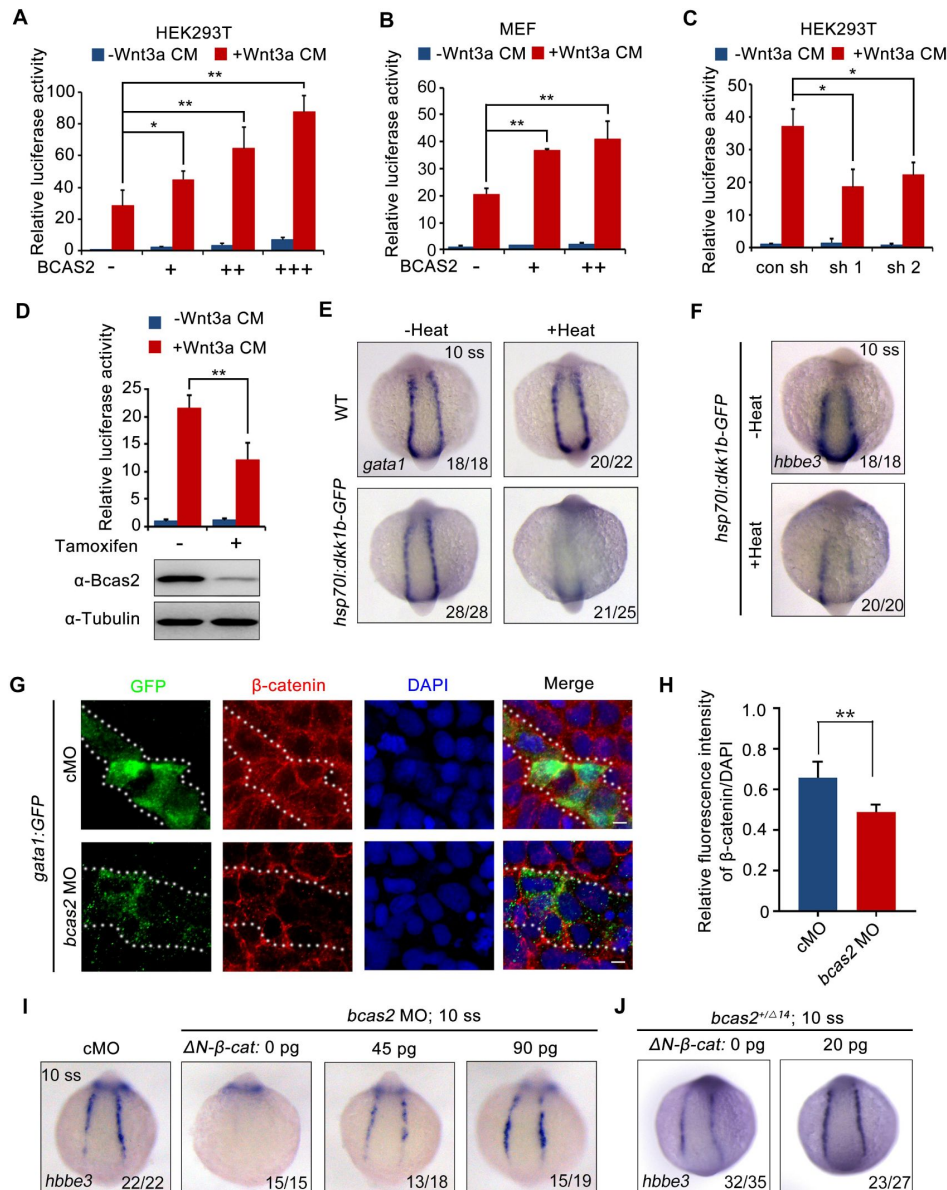


Figure 3.

BCAS2 promotes primitive hematopoiesis via activating Wnt signaling.

(A-B) Overexpression of BCAS2 increases Wnt3a-induced TOPflash activity in HEK293T cells (A) and MEFs (B). Different amounts of plasmid expressing BCAS2 (0, 80, 160, or 320 ng/well) were transfected into cells, together with the super-TOPflash luciferase and Renilla luciferase vectors. After 36 h of transfection, cells were treated with or without Wnt3a CM for 12 h and harvested for luciferase assays. * $P < 0.05$; ** $P < 0.01$ (Student's *t*-test). (C) The Wnt3a-induced TOPflash activity is decreased in BCAS2-deficient cells. HEK293T Cells were transfected with shRNA plasmids, along with indicated plasmids, and harvested for luciferase reporter assay. * $P < 0.05$ (Student's *t*-test) (D) *Bcas2*-cKO MEFs prepared from *Bcas2*^{F/F} mouse embryos were incubated in medium containing 100 μM tamoxifen for 72 h and then subjected to western blotting and luciferase reporter assay. ** $P < 0.01$ (Student's *t*-test). (E-F) Expression analysis of *gata1* (E) and *hbb33* (F) in *Tg(hsp70l:dkk1b-GFP)* embryos after heat shock at 16 hpf. (G-H) Immunofluorescence staining of β-catenin in *Tg(gata1:GFP)* embryos at 16 hpf. The embryos were injected with 8 ng of the indicated MO at the one-cell stage. The dotted lines show the GFP-positive hematopoietic progenitor cells. Scale bars, 5 μm. The relative fluorescence intensity of nuclear β-catenin was quantified in (H). ** $P < 0.01$ (Student's *t*-test). (I-J) Expression of *hbb33* in *bcas2* morphants (I) and *bcas2*^{+/Δ14} mutants (J) overexpressing Δ*N*-β-catenin. Embryos were injected with the indicated MO together with Δ*N*-β-catenin mRNA at the one-cell stage and harvested at the 10-somite stage for *in situ* hybridization.

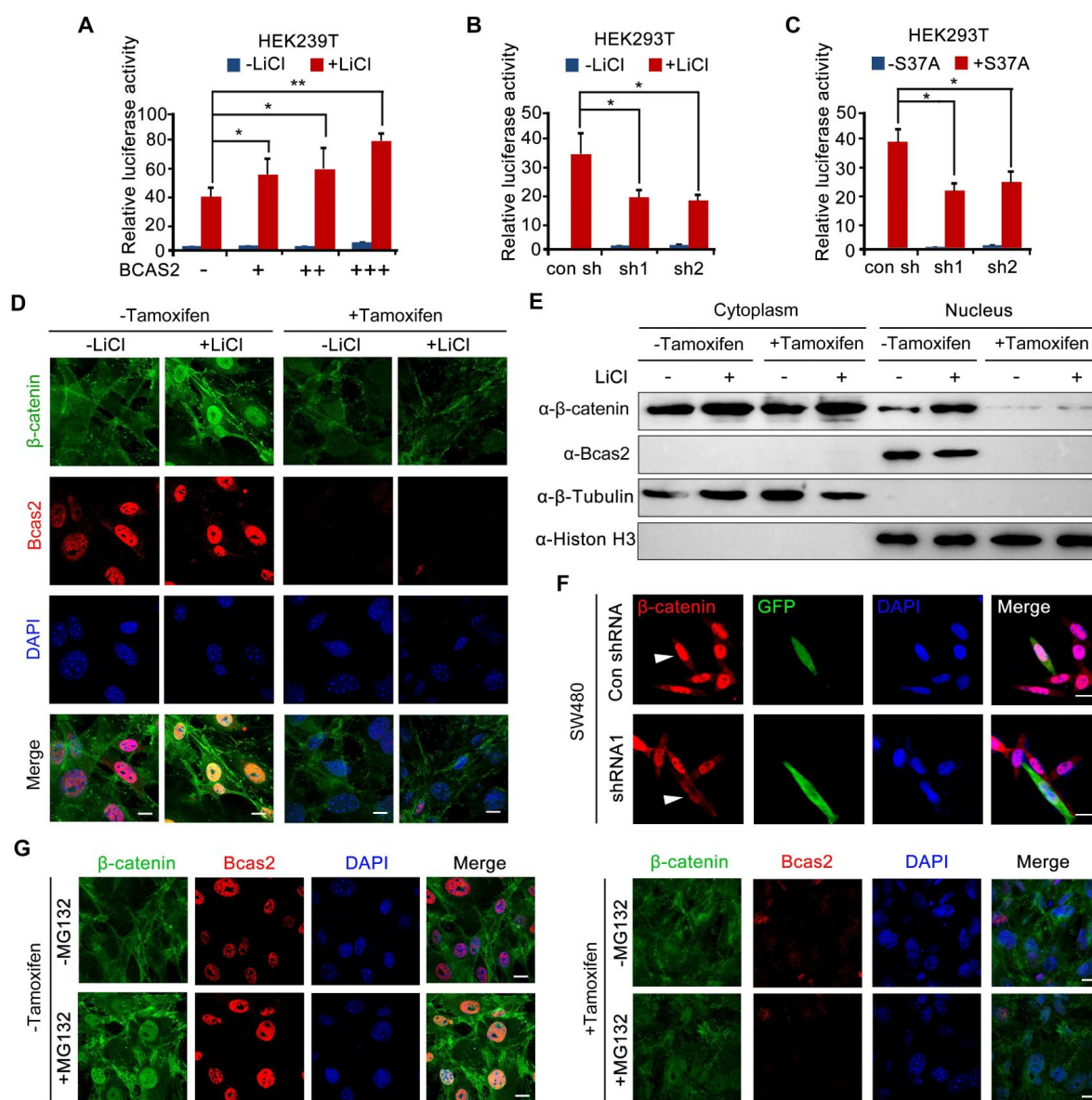


Figure 4.

BCAS2 is essential for β-catenin nuclear accumulation.

(A-C) BCAS2 enhances LiCl-induced TOPflash activity in HEK293T cells. Cells were transfected with BCAS2 expression plasmids (A), shRNA plasmids (B), or S37A-β-catenin expression plasmids (C), together with the TOPflash luciferase and Renilla luciferase vectors. After transfection, cells were subsequently treated with or without 100 ng/mL LiCl for 12 h and assayed for luciferase activity. * $P < 0.05$; ** $P < 0.01$ (Student's *t*-test). (D-E) *Bcas2*-cKO MEFs were incubated with tamoxifen for 24 h and then treated with or without 100 ng/mL LiCl. The nuclear accumulation of β-catenin was analyzed using immunofluorescence (D) and western blotting (n=3) (E) Scale bars, 10 μm. (F) SW480 cells were transfected with the indicated shRNA constructs, and the endogenous β-catenin protein was detected using immunofluorescence 48 h after transfection. The expression of GFP served as a transfection control. The arrowheads indicate the cells transfected with indicated shRNA constructs. Scale bars, 10 μm. (G) *Bcas2*-cKO MEFs were cultured in the presence of tamoxifen for 24 h and then treated with 20 μM MG132 for 6 h. The expression of BCAS2 and β-catenin was measured by immunofluorescence. Scale bars, 10 μm.

BCAS2 sequesters β -catenin within the nucleus

In addition to be affected by protein stability, the nuclear level of β -catenin is also fine-tuned by the opposing actions of nuclear export and import.^{49–51} To examine the effect of BCAS2 on the nuclear import and export of β -catenin, fluorescent recovery after photobleaching (FRAP) experiments was carried out in HeLa cells expressing GFP-tagged S37A- β -catenin. After photobleaching the nucleus, no significant difference was found in the recovery of nuclear GFP signals between the cells with and without overexpression of BCAS2 (Supplemental Figures 10A, A', and C), suggesting that BCAS2 does not regulate β -catenin nuclear import. Conversely, after photobleaching the cytoplasm, BCAS2 overexpressed cells showed a much slower recovery of cytoplasmic fluorescence (Supplemental Figures 10B, B', and C), indicating that BCAS2 inhibits β -catenin nuclear export.

It has been suggested that the nuclear exit of β -catenin can be either dependent or independent on CRM1, a major nuclear export receptor.⁵² To shed light on the mechanism underlying BCAS2 mediated β -catenin nuclear retention, we treated *Bcas2*-cKO MEFs with the CRM1-specific export inhibitor LMB.⁵³ Regardless of the presence or absence of endogenous BCAS2, LMB treatment could effectively increase the level of β -catenin in the nucleus (**Figure 5A**). Consistently, treatment of LMB was able to rescue the impaired nuclear accumulation of β -catenin in BCAS2-deficient SW480 cells (**Figure 5B**). Moreover, when *bcas2* morphants in *Tg (gata1:GFP)* background were treated with LMB from bud stage to 10 somite stage, the level of nuclear β -Catenin was partially recovered (**Figure 5C–D**). Importantly, the expression of *gata1* was also restored in *bcas2* mutants upon LMB treatment (**Figure 5E**). We further tested if BCAS2 specifically regulates CRM1-mediated nuclear export of β -catenin by analyzing the nucleocytoplasmic distribution of other known CRM1 cargoes, such as ATG3 and CDC37L.⁵⁴ Intriguingly, BCAS2 overexpression in HeLa cells only slightly enhanced the nuclear localization of CDC37L, and had no significant impact on that of ATG3 (**Supplemental Figure 11**), indicating the specificity of BCAS2-mediated inhibition of CRM1-dependent nuclear export of β -catenin. Taken together, these findings suggest that BCAS2 negatively regulates CRM1-mediated nuclear export of β -catenin.

BCAS2 directly interacts with β -catenin in the nucleus

To investigate whether BCAS2 inhibits the nuclear export of β -catenin through physical binding, HEK293T cells were transfected with Flag-tagged β -catenin and HA-tagged BCAS2 constructs. Co-immunoprecipitation (Co-IP) experiments showed that Flag- β -catenin was precipitated with HA-BCAS2 as well as endogenous BCAS2, indicating an interaction between these two proteins (**Figure 6A–B**). In addition, the interaction was enhanced upon Wnt3a stimulation (**Figure 6C**). Given that Wnt ligand stimulation ultimately induces β -catenin nuclear accumulation, this enhanced interaction implies that BCAS2 associates with β -catenin within the nucleus. Therefore, we performed the bimolecular fluorescence complementation (BiFC) assay to visualize the interaction of BCAS2 and β -catenin in living cells. In this assay, the N-terminal fragment of yellow fluorescent protein (YFP) was fused to BCAS2 (YN-BCAS2), while the C-terminal fragment was fused to β -catenin (YC- β -catenin) (**Figure 6D**). As expected, the YFP fluorescence was specifically observed in the nucleus (**Figure 6E**).

Previous studies have divided the β -catenin protein into three distinct domains, including the N-terminal domain (residues 1–133), the central domain with 12 ARM repeats (residues 134–670), and the C-terminal domain (residues 671–781).⁵⁵ To identify the BCAS2 binding site, constructs expressing various truncated forms of β -catenin were generated and co-transfected with BCAS2 into HEK293T cells (**Figure 6F**). Co-IP assays revealed that deletion of the N-terminal or C-terminal domain of β -catenin did not alter the interaction between β -catenin and BCAS2 (**Figure 6G**). In contrast, when the ARM repeats 1–12 of β -catenin were deleted, the resulting deletion

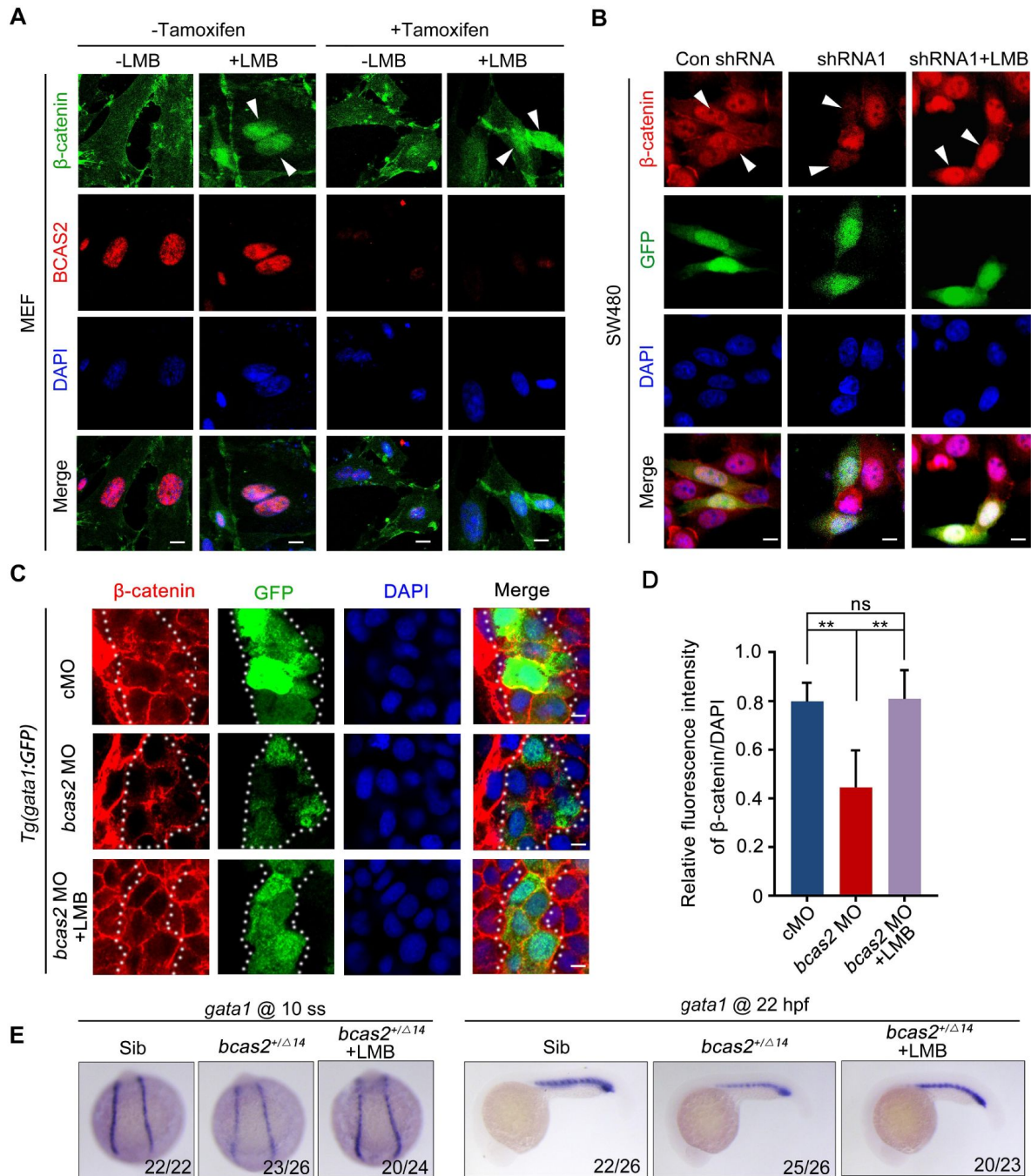


Figure 5.

BCAS2 functions in CRM1-mediated nuclear export of β-catenin.

(A) Tamoxifen-treated *Bcas2*-cKO MEFs were incubated with 20 nM LMB for 3 h. The expression of *Bcas2* and β-catenin was analyzed using immunofluorescence. The arrowheads show the cells with nuclear β-catenin accumulation. Scale bars, 10 μm. (B) SW480 cells were transfected with the indicated shRNA constructs and then treated with LMB for 3 h before immunostaining. GFP was regarded as a transfection control. The arrowheads indicate the transfected cells. Scale bars, 10 μm. (C-D) Immunofluorescence staining of β-catenin in *bcas2* morphants with *Tg(gata1:GFP)* background at 16 hpf. Embryos were exposed to 20 nM LMB from the bud stage. The dotted lines indicate the GFP-positive hematopoietic progenitor cells. Scale bars, 5 μm. The relative fluorescence intensity of nuclear β-catenin was quantified in (D). ns, not significant; ** $P < 0.01$ (Student's *t*-test). (E) *bcas2*^{+/-Δ14} embryos were treated with 20 nM LMB for 6 h and then subjected to WISH assay to analyze the expression of *gata1* at the indicated stages.

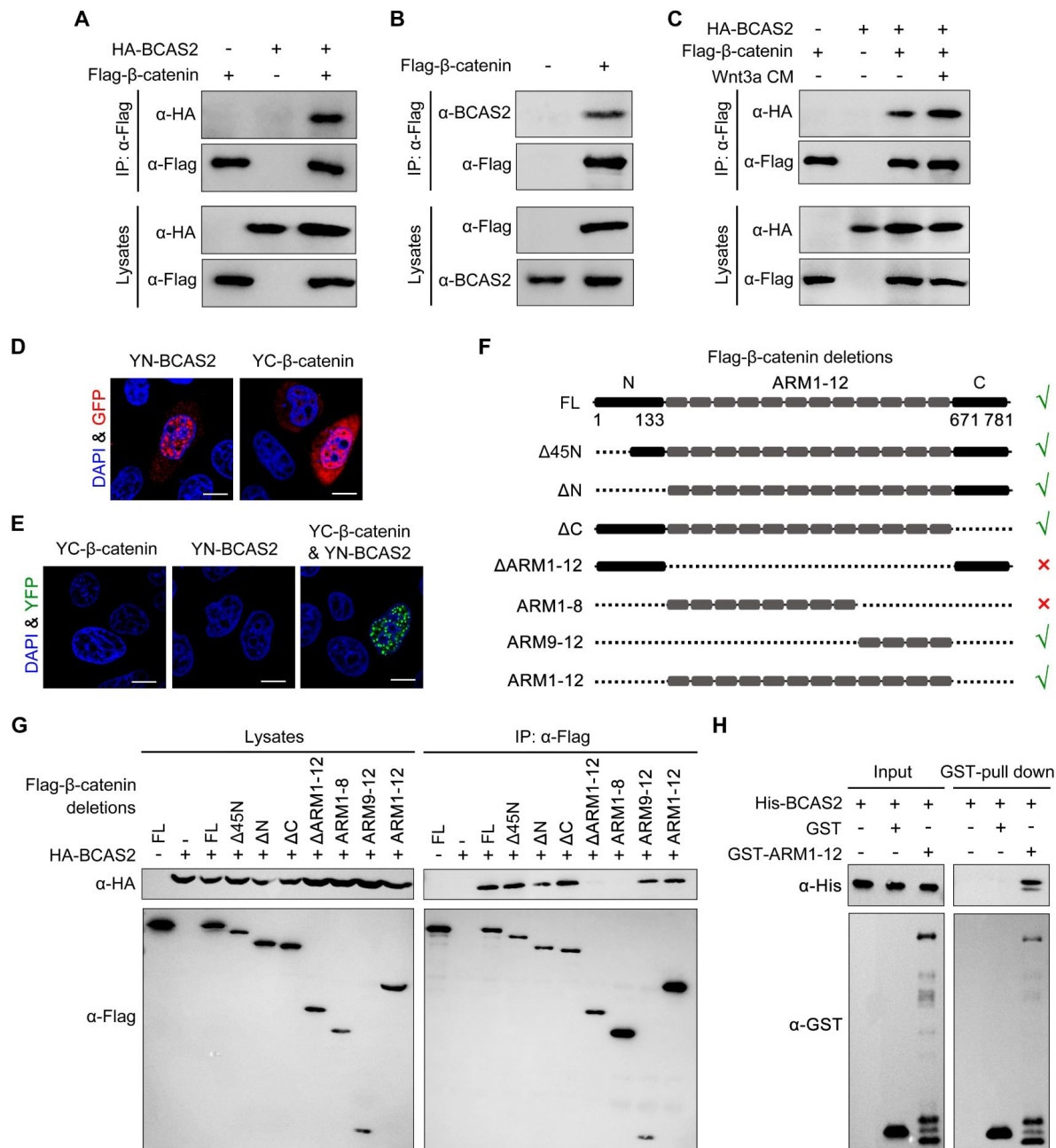


Figure 6.

BCAS2 interacts with β -catenin.

(A-C) Flag-tagged β -catenin was co-transfected with or without HA-tagged BCAS2 into HEK293T cells. Cell lysates were immunoprecipitated using anti-Flag antibody. Eluted proteins were analyzed by western blotting using indicated antibodies. In panel C, for Wnt signaling activation, cells were treated with Wnt3a CM for 5 h before harvest. (D-E) YN-BCAS2 and YC- β -catenin were either individually or collectively transfected into HeLa cells. The expression of YN-BCAS2 and YC- β -catenin was analyzed with anti-GFP antibody (D). The reconstituted YFP fluorescence in living cells was detected by confocal laser scanning microscopy with excitation at 488 nm (E). Scale bars, 10 μ m. (F) Schematics of full-length and deletion mutants of β -catenin. (G) HEK293T cells were transfected with HA-tagged BCAS2 and Flag-tagged deletion mutants of β -catenin. Cell lysates were then immunoprecipitated using anti-Flag antibody followed by western blot analysis. (H) GST pull-down assays were performed using bacterially expressed GST, GST-ARM1-12, and His-BCAS2.

mutant showed virtually no interaction with BCAS2 (**Figure 6G**). GST pull-down assay also demonstrated a direct interaction between BCAS2 and the ARM repeats of β -catenin (**Figure 6H**). These results indicate that BCAS2 physically binds to the ARM repeats of β -catenin. Furthermore, we found that the ARM repeats 9-12, but not 1-8, bound to BCAS2 (**Figure 6G**).

Transcriptional activation of the canonical Wnt target genes depends on β -catenin nuclear localization and its physical association with TCF/LEF family members. As the binding sites for TCF have been located in the ARM repeats 3-10 of β -catenin,⁵⁶ it is likely that BCAS2-mediated nuclear sequestration of β -catenin through interacting with the ARM repeats 9-12 would be compatible with the initiation of gene transcription by allowing for the association of β -catenin and TCF. To validate this possibility, co-IP assays were performed and we found that β -catenin still bound with TCF4 in the presence of BCAS2 (**Supplemental Figure 12**), confirming that the binding of BCAS2 to β -catenin would not interfere with the formation of β -catenin/TCF complex.

BCAS2 enhances β -catenin nuclear accumulation through its CC domains

To determine which domain of BCAS2 binds to β -catenin, we constructed a series of deletion mutants of BCAS2 (**Figure 7A**). Notably, we observed that among these truncated mutants, only the one lacking both CC1 and CC2 domains lost the ability to interact with β -catenin (**Figure 7B**). Moreover, these two CC domains alone or together could interact with β -catenin (**Figure 7C**). Therefore, we conclude that BCAS2 binds to β -catenin via its CC domains.

We next examined whether the CC domains are required for BCAS2 to promote Wnt/ β -catenin signaling. As shown in **Figure 7D**, overexpression of BCAS2 without the CC domains failed to increase LiCl-induced TOPflash activity in HEK-293T cells. Likewise, overexpression of the full-length or the CC domains alone, but not BCAS2 lacking the CC domains, restored the nuclear accumulation of β -catenin in *bcas2* morphants (**Figure 7E-F** and **Supplemental Figure 13A-B**). The expression of *gata1* in *bcas2* mutants was also recovered by overexpression of the full-length BCAS2, but not the truncated form without the CC-domains (**Figure 7G**). Collectively, these findings indicate that BCAS2 positively regulates Wnt signaling through sequestering β -catenin within the nucleus via its CC domains during primitive hematopoiesis.

As BCAS2 is involved in the Prp19-CDC5L spliceosome complex that regulates RNA splicing during spermiogenesis, neurogenesis, and definitive hematopoiesis,^{15,16} we wondered if this protein participates in primitive hematopoiesis via mRNA alternative splicing. To this end, we performed RNA sequencing of 10-somite stage embryos to identify abnormal events in alternative splicing in *bcas2*^{+/ Δ 14} mutants. However, upon haploinsufficiency of *bcas2*, neither the number of five major types of alternative splicing events, nor the typical forms of alternative splicing were significantly affected (**Supplemental Figure 14A-B**). Additionally, haploinsufficiency of *bcas2* did not result in the alternative splicing of *mdm4* that predisposes cells to undergo p53-mediated apoptosis in definitive hematopoiesis, as reported previously by Yu *et al.* (**Supplemental Figure 14C**).^{16,57} Furthermore, the splicing efficiency of β -catenin pre-mRNA remained almost unchanged in *bcas2*^{+/ Δ 14} mutants (**Supplemental Figure 14D**). These results demonstrate that the defects in primitive hematopoiesis of *bcas2*^{+/ Δ 14} mutants are independent of the regulatory role of Bcas2 in pre-mRNA splicing.

Discussion

BCAS2 is a 26-kDa nuclear protein involved in a multitude of developmental processes, such as Drosophila wing development, dendritic growth, and spermatogenesis.^{8,14,15,58-60} In our study, we generated *bcas2* knockout zebrafish. The heterozygotes not only showed male infertility, resembling the phenotype of *Bcas2* germ cell-specific knockout mice reported

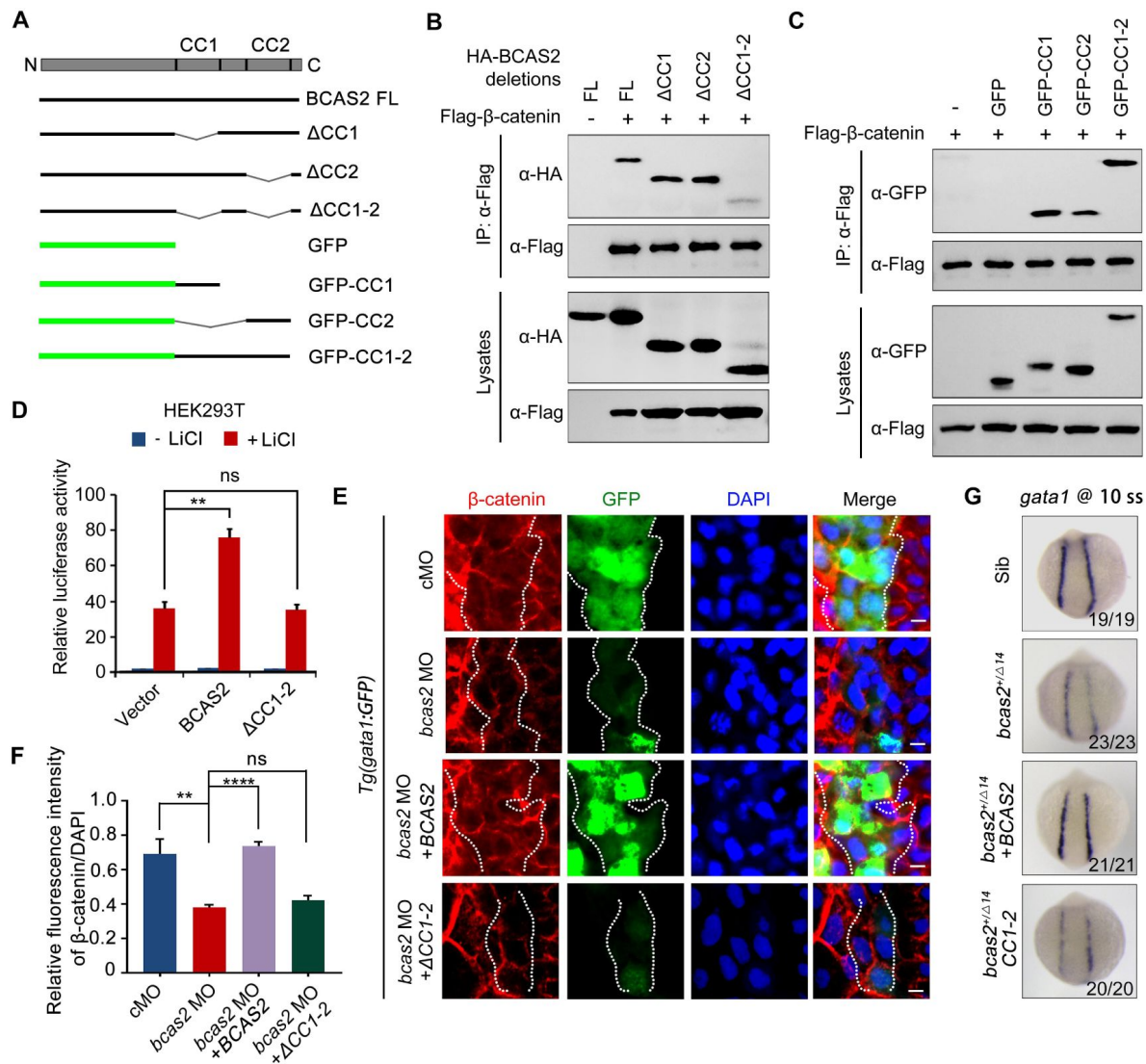


Figure 7.

BCAS2 sequesters β-catenin in the nucleus via its CC domains.

(A) Schematics of full length and deletion mutants of BCAS2. (B-C) HEK293T cells were transfected with Flag-β-catenin and indicated deletion mutants of BCAS2. Cell lysates were subjected to immunoprecipitation with anti-Flag antibody. Eluted proteins were immunoblotted using anti-HA (B) or anti-GFP antibodies (C) for BCAS2 detection. (D) HEK293T cells transfected with the indicated plasmids were treated with 100 ng/mL LiCl for 12 h, and then subjected to luciferase assay. Note that overexpression of BCAS2 without the CC domains failed to increase LiCl-induced TOPflash activity. Data represent the mean $\pm$ SD of three independent experiments. ns, not significant; ** $P < 0.01$ (Student's *t*-test). (E-F) Immunofluorescence staining of β-catenin in *Tg(gata1:GFP)* embryos at 16 hpf. The embryos were injected with 8 ng *bcas2* MO and 300 pg of full length BCAS2 mRNA or ΔCC1-2 mRNA at the one-cell stage. Scale bars, 5 μm. The relative fluorescence intensity of nuclear β-catenin was quantified in (F). ns, not significant; ** $P < 0.01$; **** $P < 0.0001$ (Student's *t*-test). (G) Transcripts of *gata1* were evaluated by WISH in *bcas2*^{+/Δ14} embryos injected with 300 pg of BCAS2 mRNA or ΔCC1-2 mRNA.

previously,¹⁵ but also exhibited impaired definitive hematopoiesis, consistent with the earlier study.¹⁶ Importantly, we found a marked decrease in the expression of the primitive erythroid progenitor markers *gata1* and *hbbe3* in these heterozygous mutants, which was rescued by overexpression of BCAS2. Moreover, the defective primitive hematopoiesis in mutant zebrafish was phenocopied in hemangioblast-specific *Bcas2* knockout mice. While the reason(s) for the discrepancy between our data and the observations made by Yu *et al.* regarding the role of *bcas2* in the development of primitive erythroid and myeloid cells remains to be determined,¹⁶ our findings in zebrafish and mouse embryos provide solid evidence that BCAS2 plays a conserved role in primitive hematopoiesis.

As demonstrated in previous studies, BCAS2 is involved in various developmental events by regulating pre-mRNA splicing.^{14–16,29,30} However, our data showed that haploinsufficiency of *bcas2* did not affect alternative splicing during primitive hematopoiesis. These results imply that one copy of the *bcas2* gene is sufficient to support mRNA splicing in zebrafish. Instead, we find that *Bcas2* promotes primitive hematopoiesis by sequestering β -catenin within the nucleus. It has been reported that the *bcas2* deletion in zebrafish embryos induces alternative splicing of *Mdm4* that predisposes cells to undergo p53-mediated apoptosis in HSPCs during definitive hematopoiesis.¹⁶ Intriguingly, we found that the loss of one copy of *bcas2* gene in zebrafish also resulted in severe impairment of HSPCs and their derivatives. It is possible that *Bcas2* might also have a role in definitive hematopoiesis independent of its splicing regulatory function.

For the past decades, given the contradictory conclusions obtained from various *in vitro* and *in vivo* studies, the function of Wnt/ β -catenin signaling in primitive hematopoiesis remains elusive and controversial.^{25–28} In the present study, we have provided several lines of evidence supporting that Wnt/ β -catenin signaling positively regulates primitive hematopoiesis: (1) Inhibition of Wnt/ β -catenin by overexpression of the canonical Wnt inhibitor Dkk1 disrupts the formation of erythrocyte progenitors at the 10-somite stage. (2) Defects in primitive hematopoiesis in *bcas2* morphants and mutants are readily restored by overexpression of ΔN - β -catenin, a constitutively active β -catenin. (3) Overexpression of the full-length BCAS2, but not the CC domain-deleted BCAS2, restores the formation of the primitive erythroid progenitor in *bcas2* mutants. (4) BCAS2 overexpression enhances the development of primitive blood cells in wild-type embryos. All these data suggest that BCAS2-mediated Wnt/ β -catenin signal activation is necessary for primitive hematopoiesis.

In addition, Wnt/ β -catenin signaling has been known as an important pathway involved in the regulation of axis determination and neural patterning during gastrulation.^{61–64} In our study, neither the heterozygous *bcas2* mutant embryos nor the very few homozygous ones, exhibited any morphological defects typically associated with inhibition of Wnt signaling, such as ventralization or brain anteriorization. This may be due to the presence of maternal *Bcas2* in heterozygous and homozygous mutant embryos which were derived from crossing *bcas2*^{+/-} males with *bcas2*^{+/-} females.

CRM1 can facilitate β -catenin nuclear export in distinct ways.⁶⁵ For example, CRM1 usually recognizes and binds with the nuclear export signal (NES) sequences in chaperon proteins, such as APC, Axin and Chibby,^{66–68} to mediate the nuclear export of β -catenin. On the other hand, CRM1 can also bind directly to and function as an efficient nuclear exporter for β -catenin.⁶⁹ Since BCAS2 has not been reported to contain any recognizable NES sequences, it will be interesting to investigate whether BCAS2 competitively inhibits β -catenin from associating with CRM1, or with the chaperone proteins.

In summary, we uncover a novel role of BCAS2 in primitive hematopoiesis through enhancing nuclear retention of β -catenin. Our study provides new insights into the mechanism of BCAS2-mediated Wnt signal activation during primitive hematopoiesis. Given that BCAS2 and Wnt

signaling are well documented to contribute to cancer development,⁷⁰⁷⁴ it is appealing to further explore whether our findings can be applied to future cancer research.

Supplemental figures

Figure S1.

Expression patterns of *bcas2* in wild-type embryos during development.

Expression of *bcas2* in wild-type embryos at the indicated developmental stages was analyzed using whole-mount *in situ* hybridization.

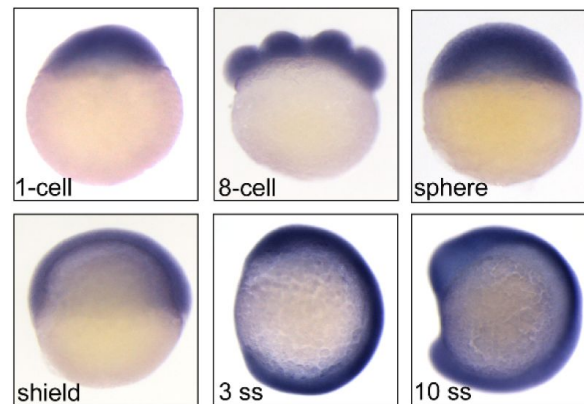
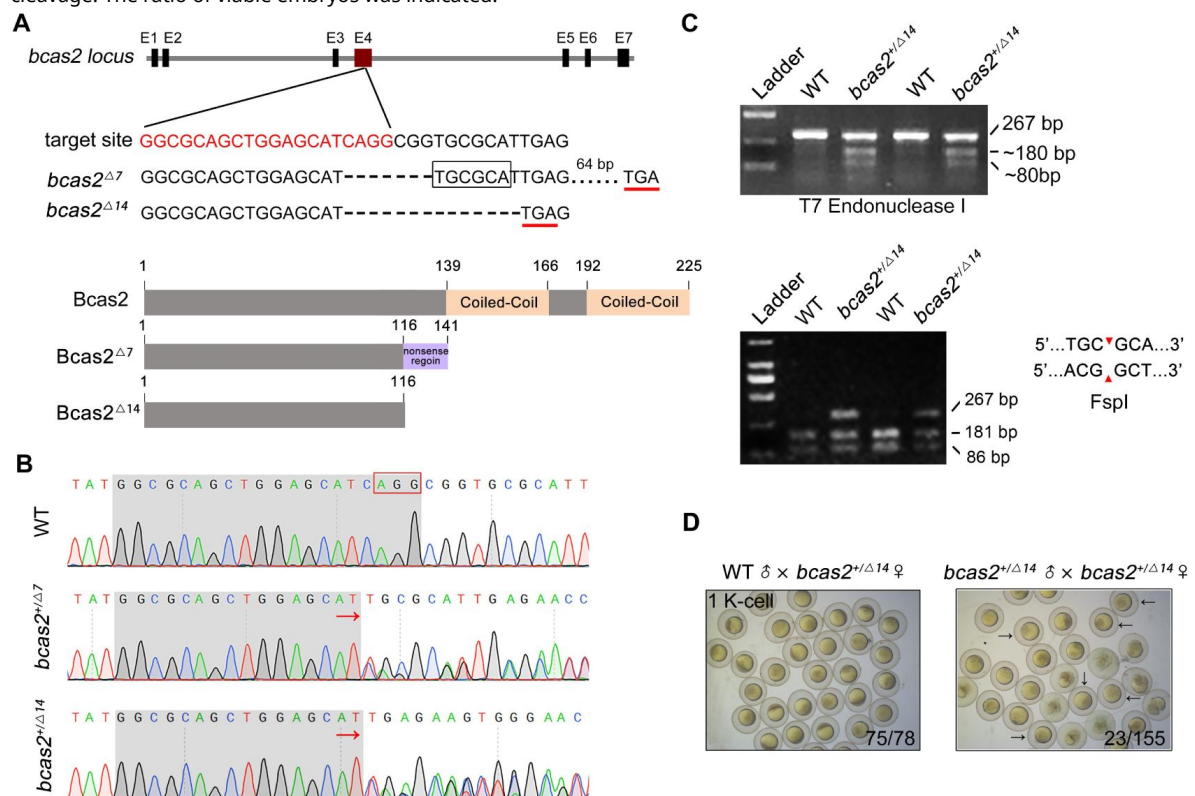


Figure S2.

Zebrafish *bcas2* mutants are generated by using CRISPR/Cas9 system.

(A) Schematic showing generation of *bcas2* mutants. Two mutant lines were obtained with mutations that resulted in premature translation termination, resulting in truncated Bcas2 proteins lacking the C-terminal CC domains. (B) Identification of *bcas2* mutations using DNA sequencing. (C) The *bcas2*^{+/Δ7} and *bcas2*^{+/Δ14} mutants were identified via T7 endonuclease (the upper panel) or FspI restriction enzyme (the lower panel) digestions. (D) Bright field images of embryos derived from crossing indicated females with heterozygous male mutants. Black arrows refer to the embryos that exhibited an abnormal cleavage. The ratio of viable embryos was indicated.



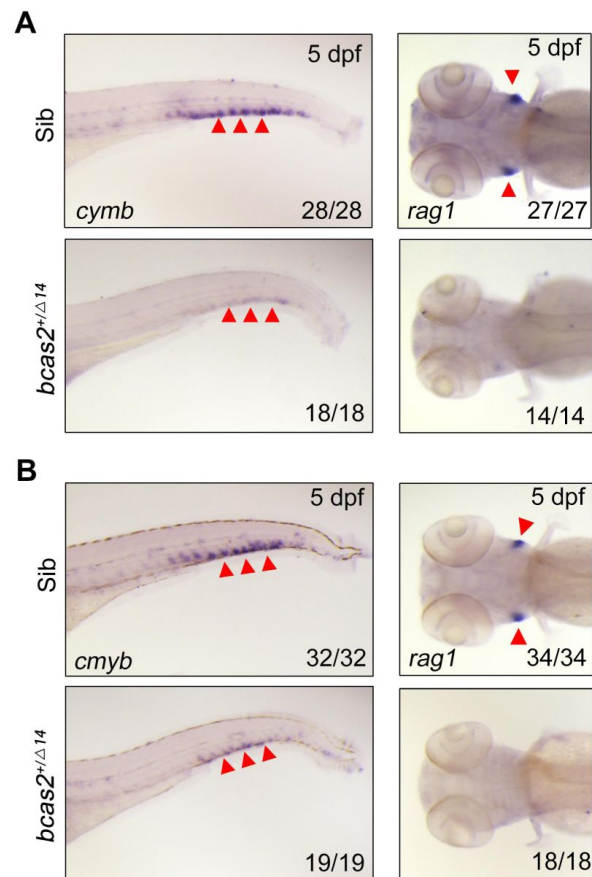


Figure S3.

***bcas2* is essential for definitive hematopoiesis.**

(A-B) Expression changes of *cymb* and *rag1* in *bcas2*^{+/Δ7} (A) and *bcas2*^{+/Δ14} (B) embryos compared with their siblings. Arrowheads on the left panels indicate the caudal hematopoietic tissue, and those on the right panels indicate the thymus.

Figure S4.

Knockdown of *bcas2* impairs primitive hematopoiesis.

(A) Western blot analysis showing the expression changes of Bcas2 protein in wild-type embryos and embryos injected with 8 ng control MO (cMO) or *bcas2* MO. (B-C) Expression of *gata1* (B) and *hbbe3* (C) in *bcas2* morphants and control embryos. (D) Detection of hemoglobin levels by α -dianisidine staining in *bcas2* morphants and control embryos.

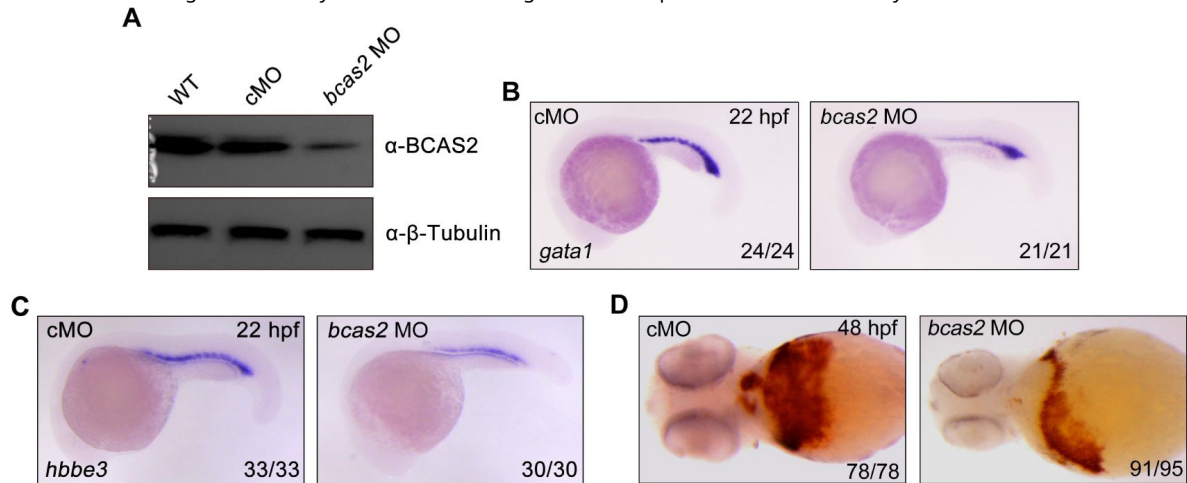


Figure S5.

***bcas2* is dispensable for the survival and proliferation of hematopoietic cells.**

(A) Cell apoptosis was detected by TUNEL assay in *bcas2*^{+/ Δ 14} embryos and their siblings with *Tg(gata1:GFP)* background. Scale bar, 50 μ m. (B-C) Cell proliferation was detected by BrdU staining in *bcas2*^{+/ Δ 14} embryos and their siblings with *Tg(gata1:GFP)* background. Scale bars, 20 μ m. The percentage of BrdU⁺/GFP⁺ cells (% over the total number of GFP⁺ cells) was quantified from six embryos in each group (C). Data represent the mean $\pm$ SD of three independent experiments. ns, not significant (Student's *t*-test).

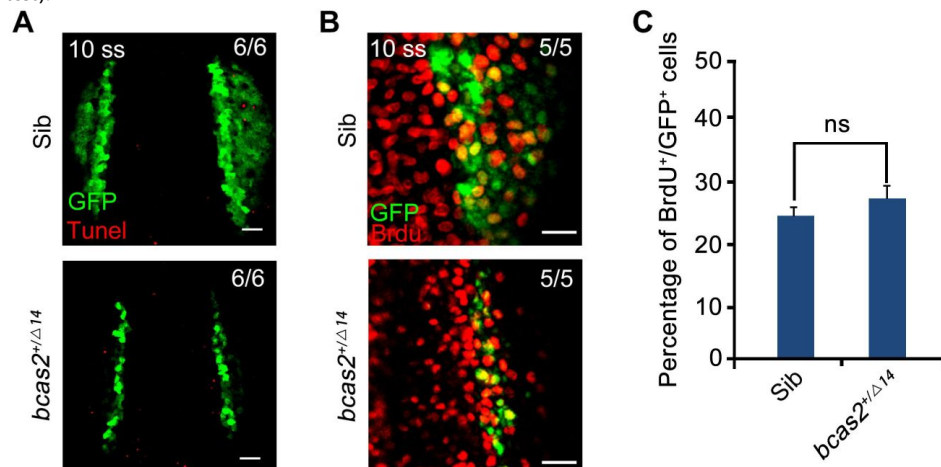


Figure S6.

Western blot analysis of HEK293T Cells transfected with corresponding shRNA constructs.

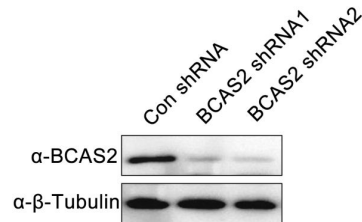


Figure S7.

Expression patterns of *cdx4* and *hoxa9a* in *bcas2*^{+/-Δ14} embryos and their siblings at the 6-somite stage. Black arrows indicate the posterior lateral mesoderm.

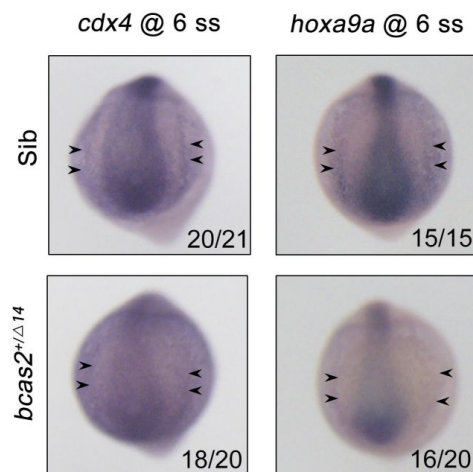


Figure S8.

Inhibition of Wnt signaling does not affect the generation of hemangioblasts or their endothelial differentiation, but impairs their hematopoietic differentiation.

Expression of hemangioblast markers *npas4l*, *scl*, *gata2* (A), endothelial marker *fli1a* (B) and erythroid progenitor marker *gata1* (C) at the indicated stages. Wild-type embryos were treated with 10 μ M CCT036477 from 9 hpf and then collected for whole-mount *in situ* hybridizations.

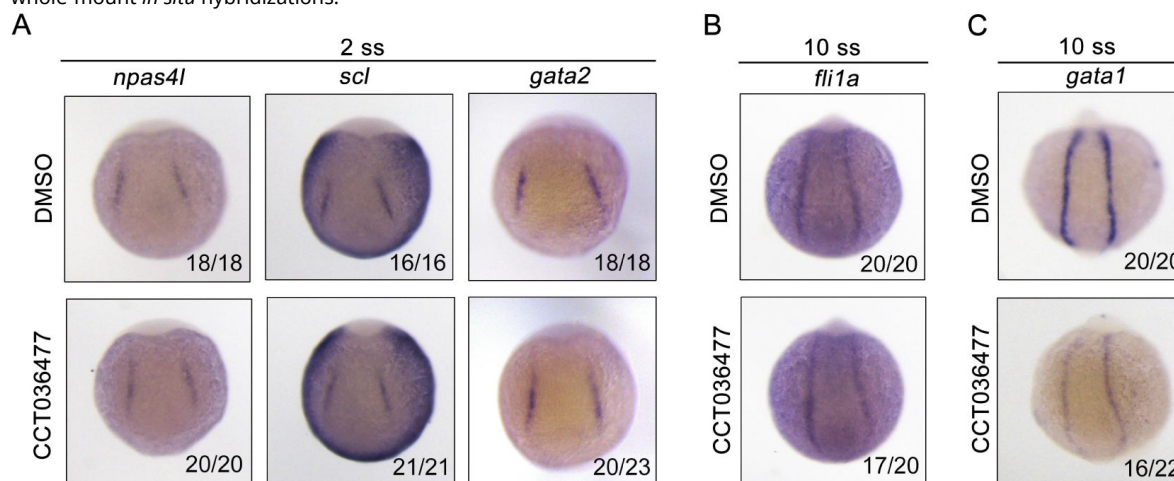
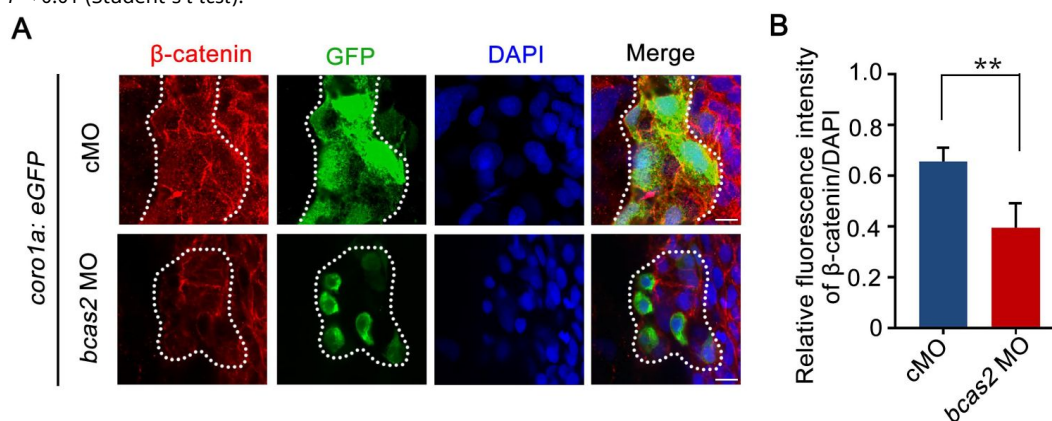


Figure S9.

Knockdown of *bcas2* significantly reduces nuclear β -catenin in the primitive myeloid cells.

Immunofluorescence staining of β -catenin in *Tg(coro1a:GFP)* embryos at 17 hpf. The embryos were injected with 8 ng of the indicated MOs at the one-cell stage and then collected for immunofluorescence staining. The dotted lines refer to the GFP-positive primitive myeloid cells. Scale bars, 10 μ m. The relative fluorescence intensity of nuclear β -catenin was quantified in (B). ** $P < 0.01$ (Student's *t*-test).



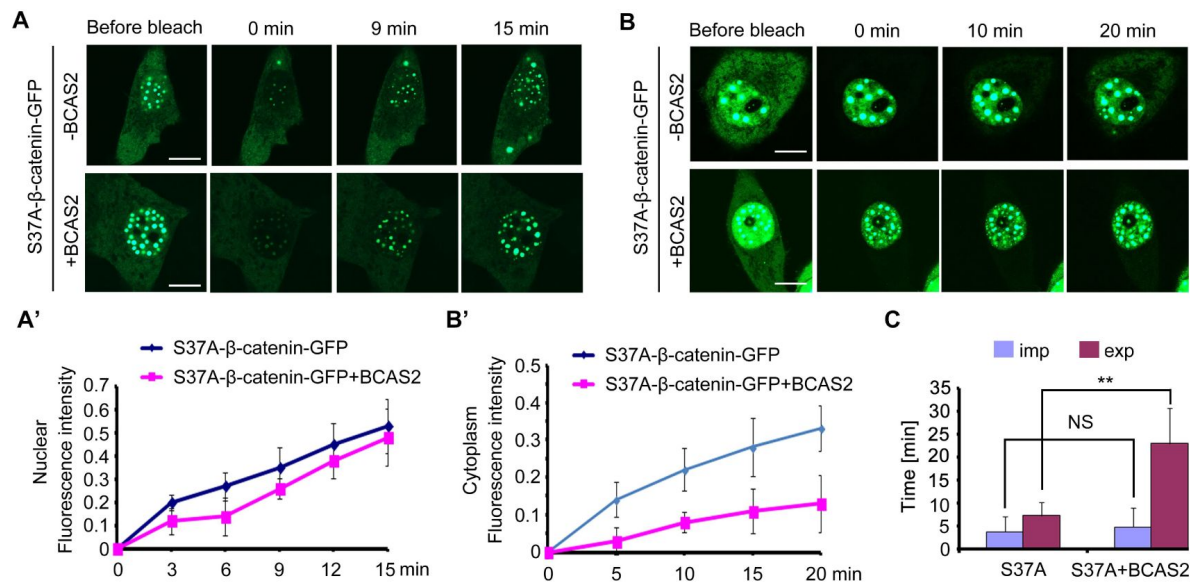


Figure S10.

BCAS2 inhibits the nuclear export of β-catenin.

(A-B') GFP-tagged S37A-β-catenin was co-expressed with or without BCAS2 in HeLa cells. The entire nucleus (A) or entire cytoplasm (B) was bleached. (A-A') Representative images at the indicated timepoints showing the kinetics of nuclear import of S37A-β-catenin are shown in (A) and the related fluorescence recovery curves are shown in (A'). Time-lapse imaging (B) and recovery curves (B') from cytoplasmic photobleaching experiments showed the kinetics of nuclear export of S37A-β-catenin. Scale bars, 10 μm. (C) Quantitative analysis of fluorescence recoveries after photobleaching (n=7). Data represent the mean ± SD of three independent experiments. Student's t-test was used. ns, not significant, **P < 0.01 (Student's t-test).

Figure S11.

Overexpression of BCAS2 slightly enhances the nuclear accumulation of CDC37L, and has no influence on the distribution of ATG3.

HeLa cells were transfected with the indicated plasmids and then subjected to immunostaining using anti-Flag and anti-HA antibodies. The nuclei were labeled with DAPI. Scale bar, 10 μ m.

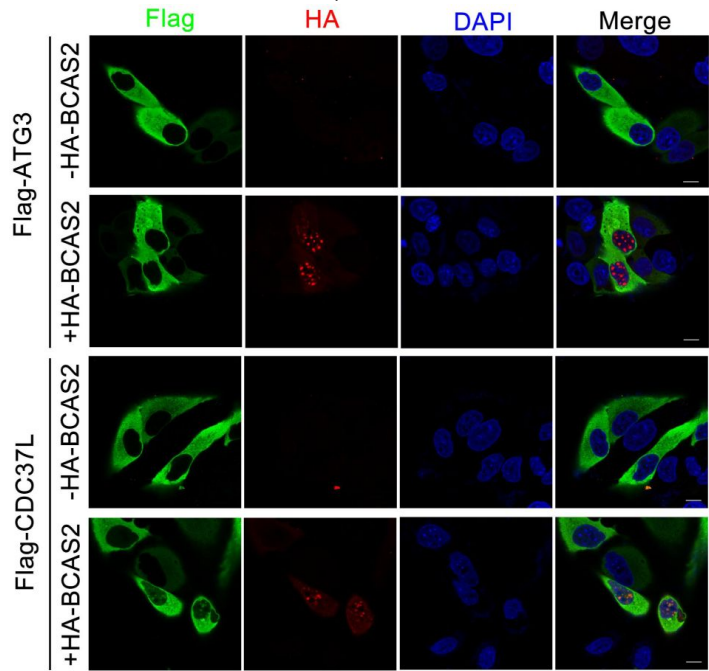
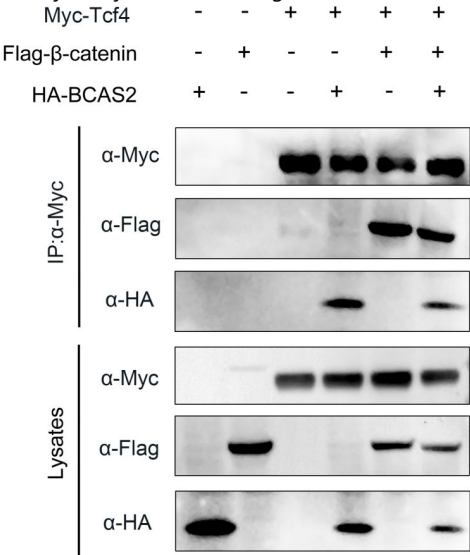


Figure S12.

The interaction between β -catenin and TCF4 remains unaffected in the presence of BCAS2.

HEK293T cells were transfected with the indicated constructs. Cell lysates were immunoprecipitated using anti-c-Myc antibody and the eluted proteins were analyzed by western blotting.



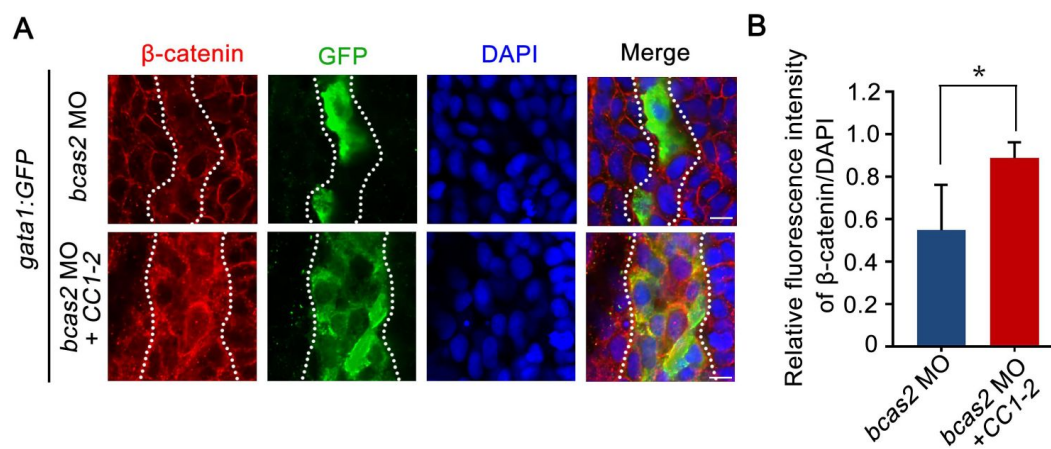


Figure S13.

Overexpression of the CC domains of BCAS2 restores nuclear β -catenin accumulation in *bcas2* morphants.

Immunofluorescence staining of β -catenin in *Tg(gata1:GFP)* embryos at 16 hpf. The embryos were injected with 8 ng *bcas2* MO and 300 pg of *BCAS2 CC1-2* mRNA at the one-cell stage. The dotted lines show the GFP-positive hematopoietic progenitor cells. Scale bars, 10 μ m. The relative fluorescence of nuclear β -catenin was quantified in (B). * $P < 0.05$ (Student's *t*-test).

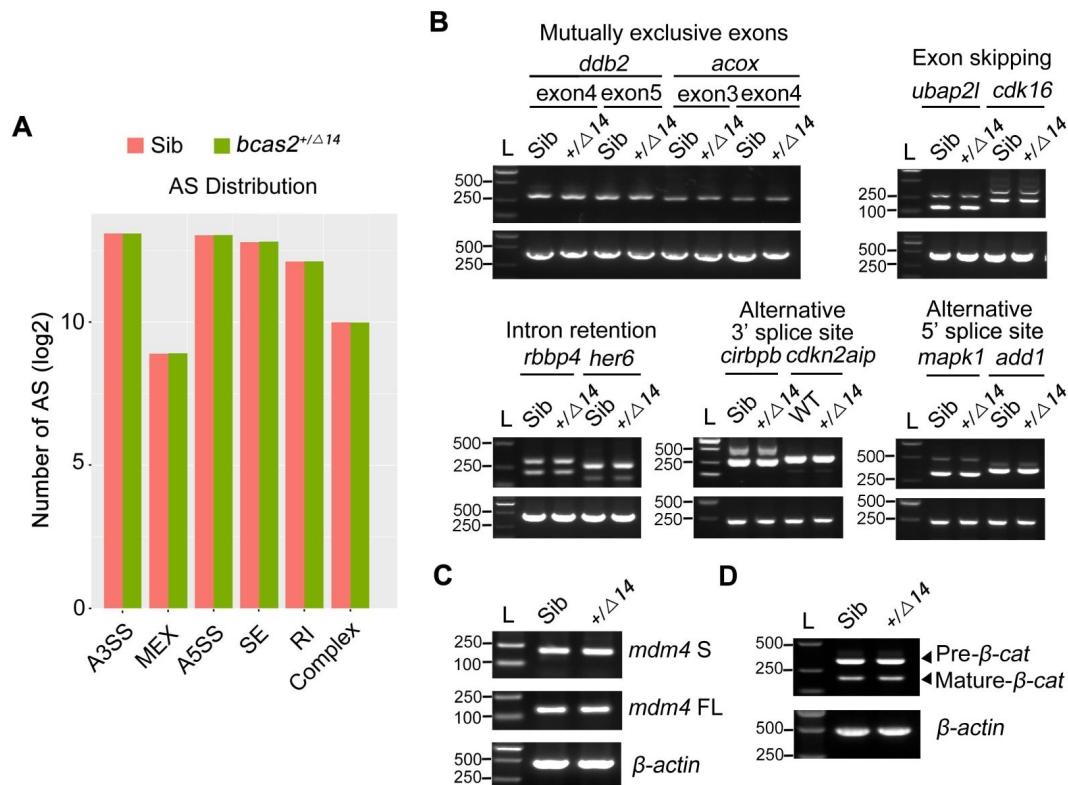


Figure S14.

Haploinsufficiency of *bcas2* does not affect pre-mRNA splicing during primitive hematopoiesis.

(A) The number of five major types of alternative splicing events was analyzed from RNA sequencing data. Embryos were lysed at the 10-somite stage and subjected to reverse transcription. The cDNA library was prepared, sequenced and then analyzed using rMATS. The five main alternative splicing types refer to exon skipping (SE), alternative 3' splicing site (A3SS), alternative 5' splicing site (A5SS), intron retention (RI), and mutually exclusive exons (MEX). (B) Examples of different types of alternative splicing were analyzed by reverse transcription PCR using total RNA from sibling and *bcas2*^{+/Δ14} embryos at the 10-somite stage. (C) Reverse transcription PCR analysis of total *mdm4*-FL and *mdm4*-S isoforms in *bcas2*^{+/Δ14} and sibling embryos at the 10-somite stage. (D) Analysis the pre-mRNA and mature mRNA of *β-catenin* in *bcas2*^{+/Δ14} and sibling embryos at the 10-somite stage.

Primers Used for Genotyping		
Strain	Directions	Sequence (5' to 3')
<i>Flk1</i> -Cre mouse	Forward	CGGTTATTCAACTTGCACCAC
	Reverse	CAGGACTGAAAGCCCAGACT
<i>Bcas2</i> ^{Flox/Flox} mouse	Forward	ATTCCAGCAGTTGGTGTGGG
	Reverse	CATTGCTGGACAGAAGGTGAG
<i>Flk1</i> -Cre; <i>Bcas2</i> ^{Flox/Flox} mouse	Forward	AGGTGTATGAATGCCTGAACAAG
	Reverse	CATTGCTGGACAGAAGGTGAG
<i>bcas2</i> knockout zebrafish	Forward	TGCACATACAGTAATAGGCTTACCC
	Reverse	GTCTGATTTGCATCAAAAGATGTGA

Supplemental Table 1

Primers Used for Genotyping

Primers Used for Reverse Transcription-PCR

Symbol	Directions	Sequence (5' to 3')
<i>ddb2</i>	Forward	CGCTCACTTAAATCTTACAAGCTGC
	Reverse	AGTCTGGGGTACGAGACAATATCT
	Reverse	TTGGCAAAGACTTGTATAACGGATC
<i>acox</i>	Forward	ATCCAGATTTC AACATGAAGACCT
	Reverse	AGGTGCCTAACATATGGAATGACTC
	Forward	TTCTCTCTCGAAGTGAGCGATATGA
	Reverse	TCTGTGCGTAGGTGCCAATAATCTC
<i>ubap2l</i>	Forward	GAATGTTGGTGTCAATGCATCAGCT
	Reverse	CTGCGTCTTTGTGTAAACAGAGCCT
<i>cdk16</i>	Forward	ATGAGGAAGATCAAACGGCAGCTGT
	Reverse	AAATTTTGCGACCGGGGTTGTTTCT
<i>rbbp4</i>	Forward	TGGGACACTCGTTCCAATAACACAT
	Reverse	CTGGAAGATTTCATCTTTGTGCGAC
<i>her6</i>	Forward	GGAGAAAAGAAGAAGAGCGAGAATC
	Reverse	GCTGCATGTTTCTGAGATGTTTCAC
<i>cirbpb</i>	Forward	TCCTACAGAGACGGTTACGACAGTT
	Reverse	CCTCACCAGGCTTGAAACTCTTACA
<i>cdkn2aip</i>	Forward	TCGCTAATGAGGACTTGTCTTTGGA
	Reverse	ATTTATCTTGTCCATCACACGCTGC
<i>mapk1</i>	Forward	GTCTGTTGGTTGTATTCTGGCTGAG
	Reverse	AGGATCATAGTACTGCTCCAGGTAC
<i>add1</i>	Forward	AGGATGCTGGATAATCTGGGCTACA
	Reverse	GTTGCGCATCTCTTGGA CTCTTTC
<i>Mdm4-FL</i>	Forward	GGTCAGGTGTCCAGTGAGTCAATAA
	Reverse	AAACCATCTGAGGAGTCTTCATCTG
<i>Mdm4-S</i>	Forward	GAAGATCCTGGTCAGACTCCTCAGA
	Reverse	CGGGAGAGAGTTGATTGGTGTGAAT
<i>β-catenin</i>	Forward	GATGACGATGTGGATAATCAGGTGC
	Reverse	GAAGTGTGTGGAAGGTATCTGCATG
	Forward	GGAGACAATTTCTCATTATTCTGGC
	Reverse	TCAGTAGTTTGGTGAGCTCTGGAAT
<i>β-actin</i>	Forward	ATGGATGATGAAATTGCCGCAC
	Reverse	ACCATCACCAGAGTCCATCACG

Supplemental Table 2

Primers Used for Reverse Transcription-PCR

Acknowledgements

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Guozhu Ning: Writing - Original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. Yu Lin: Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. Haixia Ma, Jiaqi Zhang, Liping Yang, Zhengyu Liu: Methodology, Validation. Lei Li: Resources. Xinyu He: Writing - Review & Editing. Qiang Wang: Project administration, Supervision, Funding acquisition.

Additional files

Supplemental Table 1 [↗](#)

Supplemental Table 2 [↗](#)

Supplemental Figures [↗](#)

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Reviewer #1 (Public review):

Summary:

In this manuscript, Ning et al. reported that Bcas2 played an indispensable role in zebrafish primitive hematopoiesis via sequestering β -catenin in the nucleus. The authors showed that loss of Bcas2 caused primitive hematopoietic defects in zebrafish. They unraveled that Bcas2 deficiency promoted β -catenin nuclear export via a CRM1-dependent manner in vivo and in vitro. They further validated that BCAS2 directly interacted with β -catenin in the nucleus and enhanced β -catenin accumulation through its CC domains. They unveil a novel insight into Bcas2, which is critical for zebrafish primitive hematopoiesis via regulating nuclear β -catenin stabilization rather than its canonical pre-mRNA splicing functions. Overall, the study is impressive and well-performed, although there are also some issues to address.

Strengths:

The study unveils a novel function of Bcas2, which is critical for zebrafish primitive hematopoiesis by sequestering β -catenin. The authors validated the results in vivo and in vitro. Most of the figures are clear and convincing. This study nicely complements the function of Bcas2 in primitive hematopoiesis.

Comments on revisions:

The authors have nicely answered all my questions, I have no problem.

<https://doi.org/10.7554/eLife.100497.2.sa3>

Reviewer #2 (Public review):

Summary:

Ning and colleagues present studies supporting a role for breast carcinoma amplified sequence 2 (Bcas2) in positively regulating primitive wave hematopoiesis through amplification of beta-catenin-dependent (canonical) Wnt signaling. The authors present

compelling evidence that zebrafish *bcas2* is expressed at the right time and place to be involved in primitive hematopoiesis, that there are primitive hematopoietic defects in hetero- and homozygous mutant and knockdown embryos, that Bcas2 mechanistically positively regulates canonical Wnt signaling, and that Bcas2 is required for nuclear retention of B-cat through physical interaction involving armadillo repeats 9-12 of B-cat and the coiled-coil domains of Bcas2. Overall, the data and writing are clean, clear, and compelling. This study is a first rate analysis of a strong phenotype with highly supportive mechanistic data. The findings shed light on the controversial question of whether, when, and how canonical Wnt signaling may be involved in hematopoietic development.

In the revised version of their previous work, they have included responses to some of our suggestions for minor experiments and edits. We previously suggested they examine the structural compatibility of a Bcas2/beta-catenin dimer with binding to the DNA-binding protein Tcf7l1 (previously Tcf3), which would be expected for a beta-catenin nuclear-retention factor that potentiates canonical Wnt signaling responses. Although the authors did not test compatibility of Bcas2 with Tcf3 binding to beta-catenin, they show that a three-way complex with the family member Tcf4 is possible (Fig. S12), which suggests that Lef/Tcf family binding in general is plausible.

The authors' acceptance of our suggestion to evaluate *cdx* and *hox* gene expression is welcome, as these genes have previously been defined as canonical Wnt targets (Lengerke et al., 2009) that regionalize the lateral plate mesoderm (LPM) and confer pre-hematopoietic identity there (Davidson et al., 2003; Davidson and Zon, 2004). The authors' finding that *cdx4* and *hoxa9a* are diminished in the *bcas2* mutants (Fig. S7) validates this suggestion and seem to imply that the primary defect here is specification of the early hematopoietic field in the LPM, however the results are a little confusing or surprising given that *scl* - which is unaffected in the *bcas2* mutant (Fig. 2A) - is a downstream target of Cdx4 (Davidson et al., 2003, Fig. 1b, 3d). The results in the current submission imply that early maintenance of pre-hematopoietic competence in the LPM is a canonical-Wnt-directed phenomenon separable from the earliest specification of the hematopoietic field. We believe it would be of value to further evaluate regulation of *cdx1*, which has been shown to cooperate with *cdx4* in regulation of the LPM hematopoietic field, as well as analyze some of the putative downstream *hox* family targets.

We previously reviewed the article as suitable for publication and we continue to support our prior assessment. The authors have presented strong data supporting a role for Bcas2 in hematopoietic development across phyla and a mechanistic involvement in promoting canonical Wnt signaling.

Strengths:

- (1) The study features clear and compelling phenotypes and results.
- (2) The manuscript narrative exposition and writing are clear and compelling.
- (3) The authors have attended to important technical nuances sometimes overlooked, for example, focusing on different pools of cytosolic or nuclear b-catenin.
- (4) The study sheds light on a controversial subject: regulation of hematopoietic development by canonical Wnt signaling and presents clear evidence of a role.
- (5) The authors present evidence of phylogenetic conservation of the pathway.

<https://doi.org/10.7554/eLife.100497.2.sa2>

Reviewer #3 (Public review):

Summary:

This manuscript utilized zebrafish *bcas2* mutants to study the role of *bcas2* in primitive hematopoiesis, and further confirms that it has a similar function in mice. Moreover, they showed that *bcas2* regulates the transition of hematopoietic differentiation from angioblasts via activating Wnt signaling. By performing a series of biochemical experiments, they also showed that *bcas2* accomplishes this by sequestering β -catenin within the nucleus, rather than through its known function in pre-mRNA splicing.

Strengths:

The work is well-performed, and the manuscript is well-written.

Comments on revisions:

The revised manuscript is substantially improved, and all my previous questions are now well addressed.

<https://doi.org/10.7554/eLife.100497.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public Review):

Summary:

In this manuscript, Ning et al. reported that Bcas2 played an indispensable role in zebrafish primitive hematopoiesis via sequestering β -catenin in the nucleus. The authors showed that loss of Bcas2 caused primitive hematopoietic defects in zebrafish. They unraveled that Bcas2 deficiency promoted β -catenin nuclear export via a CRM1-dependent manner in vivo and in vitro. They further validated that BCAS2 directly interacted with β -catenin in the nucleus and enhanced β -catenin accumulation through its CC domains. They unveil a novel insight into Bcas2, which is critical for zebrafish primitive hematopoiesis via regulating nuclear β -catenin stabilization rather than its canonical pre-mRNA splicing functions. Overall, the study is impressive and well-performed, although there are also some issues to address.

Strengths:

The study unveils a novel function of Bcas2, which is critical for zebrafish primitive hematopoiesis by sequestering β -catenin. The authors validated the results in vivo and in vitro. Most of the figures are clear and convincing. This study nicely complements the function of Bcas2 in primitive hematopoiesis.

Weaknesses:

A portion of the figures were over-exposed.

Thank you for the time reviewing our manuscript. We agree with your suggestion and the exposure of Figure 5C and Figure 7E has been reduced. We hope that the revisions will meet your expectation.

Reviewer #2 (Public Review):

Summary:

Ning and colleagues present studies supporting a role for breast carcinoma amplified sequence 2 (Bcas2) in positively regulating primitive wave hematopoiesis through amplification of beta-catenin-dependent (canonical) Wnt signaling. The authors present compelling evidence that zebrafish *bcas2* is expressed at the right time and place to be involved in primitive hematopoiesis, that there are primitive hematopoietic defects in hetero- and homozygous mutant and knockdown embryos, that Bcas2 mechanistically positively regulates canonical Wnt signaling, and that Bcas2 is required for nuclear retention of B-cat through physical interaction involving armadillo repeats 9-12 of B-cat and the coiled-coil domains of Bcas2. Overall, the data and writing are clean, clear, and compelling. This study is a first-rate analysis of a strong phenotype with highly supportive mechanistic data. The findings shed light on the controversial question of whether, when, and how canonical Wnt signaling may be involved in hematopoietic development. We detail some minor concerns and questions below, which if answered, we believe would strengthen the overall story and resolve some puzzling features of the phenotype. Notwithstanding these minor concerns, we believe this is an exceptionally well-executed and interesting manuscript that is likely suitable for publication with minor additional experimental detail and commentary.

Strengths:

- (1) The study features clear and compelling phenotypes and results.
- (2) The manuscript narrative exposition and writing are clear and compelling.
- (3) The authors have attended to important technical nuances sometimes overlooked, for example, focusing on different pools of cytosolic or nuclear b-catenin.
- (4) The study sheds light on a controversial subject: regulation of hematopoietic development by canonical Wnt signaling and presents clear evidence of a role.
- (5) The authors present evidence of phylogenetic conservation of the pathway.

Weaknesses:

- (1) The authors present compelling data that Bcas2 regulates nuclear retention of B-cat through physical association involving binding between the Bcas2 CC domains and B-cat arm repeats 9-12. Transcriptional activation of Wnt target genes by B-cat requires physical association between B-cat and Tcf/Lef family DNA binding factors involving key interactions in Arm repeats 2-9 (Graham et al., Cell 2000). Mutually exclusive binding by B-cat regulatory factors, such as ICAT that prevent Tcf-binding is a documented mechanism (e.g. Graham et al., Mol Cell 2002). It would appear - based on the arm repeat usage by Bcas2 (repeats 9-12)-that Bcas2 and Tcf binding might not be mutually exclusive, which would support their model that Bcas2 physical association with B-cat to retain it in the nucleus would be compatible with co-activation of genes by allowing association with Tcf. It might be nice to attempt a three-way co-IP of these factors showing that B-cat can still bind Tcf in the presence of Bcas2, or at least speculate on the plausibility of the three-way interaction.

We appreciate your assessment and generous comments for the manuscript. As you mentioned, the binding sites for TCF on β -catenin almost do not overlap with those for BCAS2. It is likely that BCAS2-mediated nuclear sequestration of β -catenin would be compatible with the initiation of gene transcription by allowing TCF to associate with β -catenin. To test this possibility, we have taken your suggestion and performed co-IP assays. The results showed that β -catenin still bound with TCF4 in the presence of BCAS2 (Supplemental Figure 12), confirming that the binding of BCAS2 to β -catenin would not interfere with the formation of β -catenin/TCF complex.

(2) A major way that canonical Wnt signaling regulates hematopoietic development is through regulation of the LPM hematopoietic competence territories by activating expression of *cdx1a*, *cdx4*, and their downstream targets *hoxb5a* and *hoxa9a* (Davidson et al., Nature 2003; Davidson et al., Dev Biol 2006; Pilon et al., Dev Biol 2006; Wang et al., PNAS 2008). Could the authors assess (in situ) the expression of *cdx1a*, *cdx4*, *hoxb5a*, and *hoxa9a* in the *bcas2* mutants?

We agree with your suggestion and have examined the expression of *cdx4* and *hoxa9a* by performing WISH. Diminished expression of *cdx4* and *hoxa9a* was detected in the lateral plate mesoderm of *bcas2*^{+/-} embryos at the 6-somite stage (Supplemental Figure 7).

(3) The authors show compellingly that even heterozygous loss of *bcas2* has strong Wnt-inhibitory effects. If *Bcas2* is required for canonical Wnt signaling and *bcas2* is expressed ubiquitously from the 1-cell stage through at least the beginning of gastrulation, why do *bcas2* KO embryos not have morphological axis specification defects consistent with loss of early Wnt signaling, like loss of head (early), or brain anteriorization (later)? Could the authors provide some comments on this puzzle? Or if they do see any canonical Wnt signaling patterning defects in het- or homozygous embryos, could they describe and/or present them?

You have raised an interesting question. In fact, we did not observe ventralization or axis determination defects in the early embryos of *bcas2*^{+/-} mutants. Even in the very small number of homozygous mutant embryos, we did not find such morphological defects. Given that the homozygous and heterozygous mutant embryos were derived from crossing *bcas2*^{+/-} males with *bcas2*^{+/-} females, maternal *Bcas2* might still remain and function in these embryos during gastrulation when axis determination and neural patterning took place. Accordingly, we have expanded our discussion to incorporate these insights (Line 565-572).

Reviewer #3 (Public Review):

Summary:

This manuscript utilized zebrafish *bcas2* mutants to study the role of *bcas2* in primitive hematopoiesis and further confirms that it has a similar function in mice. Moreover, they showed that *bcas2* regulates the transition of hematopoietic differentiation from angioblasts via activating Wnt signaling. By performing a series of biochemical experiments, they also showed that *bcas2* accomplishes this by sequestering b-catenin within the nucleus, rather than through its known function in pre-mRNA splicing.

Strengths:

The work is well-performed, and the manuscript is well-written.

Weaknesses:

Several issues need to be clarified.

(1) Is wnt signaling also required during hematopoietic differentiation from angioblasts? Can the authors test angioblast and endothelial markers in embryos with wnt inhibition? Also, can the authors add export inhibitor LMB to the mouse mutants to test if sequestering of b-catenin by *bcas2* is conserved during primitive hematopoiesis in mice?

Thank you very much for your appreciation and detailed assessment. To test whether Wnt signaling is also required during hematopoietic differentiation from angioblasts, wild-type embryos were exposed to 10 μM CCT036477, a small molecule β-catenin antagonist, from 9

hpf and then collected for WISH experiments. As shown in Supplemental Figure 8, the expression of hemangioblast markers *npas4l*, *scl*, and *gata2* and endothelial marker *fli1a* remained unchanged, but the expression of erythroid progenitor marker *gata1* was significantly reduced. These results suggest that canonical Wnt pathway may not be required for the generation of hemangioblasts or their endothelial differentiation, but is pivotal for their hematopoietic differentiation.

It is quite difficult to validate the conserve role of BCAS2 during primitive hematopoiesis in mice, because the toxicity of LMB may cause severe adverse effects in mice.[1,2]

(2) Bcas2 is required for primitive myelopoiesis in ALM. Does bcas2 play a similar function in primitive myelopoiesis, or is bcas2/b-catenin interaction more important for hematopoietic differentiation in PLM?

You have raised an important question. In our study, we have demonstrated that the expression of myeloid progenitor marker *pu.1* was significantly decreased in *bcas2* mutants, hinting that Bcas2 is pivotal for primitive myelopoiesis. To further clarify the function of Bcas2 in primitive myelopoiesis, we injected 8 ng of *bcas2* morpholino into *Tg(coro1a:GFP)* embryos at the 1-cell stage and examined β -catenin distribution at 17 hpf via immunostaining. We observed a significant decline of nuclear β -catenin in primitive myeloid cells (Supplemental Figure 9), indicating that Bcas2 is highly likely to play a similar role in sequestering β -catenin within the nucleus during primitive myelopoiesis.

(3) Is it possible that CC1-2 fragment sequester b-catenin? The different phenotypes between this manuscript and the previous article (Yu, 2019) may be due to different mutations in bcas2. Is it possible that the bcas2 mutation in Yu's article produces a complete CC1-2 fragment, which might sequester b-catenin?

This is an interesting perspective. To test the possibility that CC1-2 sequesters β -catenin, mRNA expressing the CC domains of BCAS2 has been co-injected with *bcas2* morpholino into *Tg(gata1:GFP)* embryo at the one-cell stage. Increased nuclear β -catenin levels were detected in the GFP-positive hematopoietic progenitor cells at 16 hpf (Supplemental Figure 11). Our findings support that CC1-2 fragment of BCAS2 can sequester β -catenin within the nucleus.

In the previous article (Yu, 2019), a deletion 5 bases mutation in the third exon of BCAS2 was produced by TALEN, therefore the CC domains of this mutant should be affected. It is difficult to conclude that the mutant BCAS2 protein in Yu's study still remains association with β -catenin.

(4) Can the author clarify what embryos the arrows point to in SI Figure 2D? In SI Figure 6B and B', can the author clarify how the nucleus and cytoplasm are bleached? In B, the nucleus also appears to be bleached.

Thank you for your query and suggestion. In our revisions, the corresponding clarifications have been supplemented (Line 239-242; Line 978-979).

We acknowledge that the nuclei in both the BCAS2 overexpression group and control group were slightly bleached. Given that we have performed real-time analysis for fluorescent recovery after photobleaching, and we have observed a much slower recovery of cytoplasmic fluorescence in BCAS2 overexpressed cells, the conclusion that BCAS2 inhibits the nuclear export of β -catenin but not its nuclear import, remains changed.

Reviewer #1 (Recommendations For The Authors):

Major concerns:

*(1) In this study, the authors detected β -catenin distribution in erythrocytes (*gata1-GFP*⁺ cells). Estimating the β -catenin distribution in the myeloid cells is recommended.*

Thank you for your assessment and we have taken your suggestion. *Tg(coro1a:GFP)* embryos, which is commonly used to track both macrophages and neutrophils,[3] were injected with 8 ng of *bcas2* morpholino into at the 1-cell stage and collected for immunostaining to examine the β -catenin distribution at 17 hpf. We observed a significant decline of nuclear β -catenin in primitive myeloid cells (Supplemental Figure 9). This result indicates that *Bcas2* is highly likely to play a similar role in sequestering β -catenin within the nucleus during primitive myelopoiesis.

(2) The reduced nuclear localization of β -catenin in Figure 3H required further evidence. It would be helpful if the authors quantified the fluorescence intensity in the cell nucleus and cytoplasm. Meanwhile, the figures (Figure 5C, Figure 7E) were over-exposed. Please validate these figures.

Thank you for your suggestions. We agree with you that the fluorescence intensity of β -catenin in the nucleus and cytoplasm should be quantified. However, as the nucleus comprises a large part of the cell, we believe it would be more appropriate to quantify the relative fluorescence intensity by dividing the fluorescence intensity of nuclear β -catenin by the fluorescence intensity of DAPI.

Such quantifications have been added for Figure 3G, 5C, 7E, S9A, and S13A. In addition, we have reduced the exposure of Figure 5C and Figure 7E. We hope that you will be satisfied with the revisions.

*(3) The authors used cKO mice to validate that the erythrocytes were eliminated. It would be interesting to detect β -catenin distribution by immunofluorescent staining in primitive hematopoietic cells in cKO mice. Addressing this issue can provide further evidence to support the conservation of *Bcas2*.*

We appreciate your suggestion. However, we found that red blood cells were almost eliminated in the yolk sac of *Bcas2*^{F/F};*Flk1-Cre* mice at E12.5. It is difficult to further detect β -catenin distribution in primitive erythroid cells in these mice.

*(4) The authors discovered that *Bcas2* mediated β -catenin nuclear export in a CRM1-dependent manner. CRM1 is a key regulator involved in the majority of factors of nuclear export via recognizing specific nuclear export signals (NES). Validating the NES of *Bcas2* is recommended. Furthermore, I wonder about the relationship between *Bcas2* and CRM1 in regulating β -catenin nuclear export. One possibility is that *Bcas2* covers the NES to inhibit the interaction between CRM1 and β -catenin, thus leading to β -catenin accumulation in the cell nucleus. The authors should discuss this possibility accordingly.*

Thank you for providing an interesting perspective. CRM1-mediated nuclear export of β -catenin usually requires CRM1 recognition and binding with the NES sequences in chaperon proteins, such as APC, Axin and Chibby.[4-6] Moreover, CRM1 can bind directly to and function as an efficient nuclear exporter for β -catenin.[7] Since BCAS2 has not been reported to contain any recognizable NES sequences, it will be interesting to investigate whether BCAS2 competitively inhibits β -catenin from associating with CRM1, or with the chaperone proteins. We have rewritten the discussion on CRM1-dependent nuclear export of β -catenin in line with your comments (Line 572-578).

*(5) It would be interesting if the authors could answer the specificity in *Bcas2*-mediated protein nuclear export pathway. The authors should detect other classical factors (CRM1*

mediated) distribution when loss of Bcas2.

Thank you for bringing up this point. To test whether BCAS2 specifically regulates CRM1-mediated nuclear export of β -catenin, we have investigated the nucleocytoplasmic distribution of other known CRM1 cargoes, such as ATG3 and CDC37L.[8] BCAS2 overexpression in HeLa cells slightly enhanced the nuclear localization of CDC37L, and had no significant impact on that of ATG3 (Supplemental Figure 11), indicating the specificity of BCAS2 in the regulation of CRM1-dependent nuclear export of β -catenin.

Minor concerns:

(1) The name "bcas2 Δ 7+/- and bcas2 Δ 14+/-" should be changed into "bcas2+/ Δ 7 and bcas2+/ Δ 14"(+/ Δ 7 or +/ Δ 14 should be superior on the right).

Thank you for your suggestion. We have changed the names of the mutants throughout the manuscript.

(2) The scale bar position in the figures should be unified.

We agree with your suggestion and have unified the scale bar position in all figures.

(3) In Figure 4E, "Nuclear" should be changed into "Nucleus".

We apologize for the mistake and Figure 4E has been revised.

(4) There are some unaesthetic issues in the figures. The figures need to be further edited. Figure 3H " β -catenin and Merge", Figure 4D "Merge". All these words should be centered in the figures.

Thank you. We have edited all the figures to ensure that the text is centered.

Reviewer #2 (Recommendations For The Authors):

(1) It would be nice to have whole blot images for the Westerns in Supplementary Info.

Thank you for your suggestion. Whole images for immunoblotting have been supplemented as Source data.

(2) Line 292 change 5 hpf to 5 dpf.

(3) Line 301 change "primary" to "primitive"?

We apologize for the mistakes. We have incorporated these suggestions in the revised manuscript and reexamined spelling throughout the paper.

(4) Figure S2C: is "Maker" a typographical error? Change to "ladder"?

We apologize for this typographical error and we have revised it in Figure S2C.

Reference

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